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Review

A Review of Brain–Computer Interface-Based Language Decoding: From Signal Interpretation to Intelligent Communication

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Abstract: Brain–computer interface (BCI) technologies for language decoding have emerged as a transformative bridge between neuroscience and artificial intelligence (AI), enabling direct neural–computational communication. The current literature provides detailed insights into individual components of BCI systems, from neural encoding mechanisms to language decoding paradigms and clinical applications. However, a comprehensive perspective that captures the parallel evolution of cognitive understanding and technological advancement in BCI-based language decoding remains notably absent. Here, we propose the Interpretation–Communication–Interaction (ICI) architecture, a novel three-stage perspective that provides an analytical lens for examining BCI-based language decoding development. Our analysis reveals the field's evolution from basic signal interpretation through dynamic communication to intelligent interaction, marked by three key transitions: from single-channel to multimodal processing, from traditional pattern recognition to deep learning architectures, and from generic systems to personalized platforms. This review establishes that BCI-based language decoding has achieved substantial improvements in regard to system accuracy, latency reduction, stability, and user adaptability. The proposed ICI architecture bridges the gap between cognitive neuroscience and computational methodologies, providing a unified perspective for understanding BCI evolution. These insights offer valuable guidance for future innovations in regard to neural language decoding technologies and their practical application in clinical and assistive contexts.

Keywords: BCI; language decoding; neural signal processing; deep learning; human–AI interaction

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1. Introduction

Language processing, as a fundamental component of human higher cognition, serves as the cornerstone of complex thought and interpersonal communication [1,2]. The recent integration of neuroscience and artificial intelligence (AI) has transformed our understanding of neural language implementation, revealing remarkable parallels between human neural representations and computational models in language processing [3–5]. This scientific advancement has catalyzed the emergence of brain–computer interface (BCI) technologies for language decoding, establishing direct communication pathways between the central nervous system and computational systems [6–8].

The evolution of BCI-based language decoding has demonstrated remarkable progress in terms of both technological advancement and practical applications, particularly through the integration of linguistic theories and cognitive frameworks. The synergy between neuroscience, computational linguistics, and artificial intelligence has fundamentally transformed our understanding of neural language processing mechanisms, enabling more sophisticated and naturalistic decoding approaches. This integration has particularly benefited from advances in cognitive neuroscience and computational linguistics, leading to more robust and adaptable systems capable of handling complex language tasks. Technically, the field has transitioned from single-signal decoding to sophisticated multimodal integration, incorporating diverse neural and physiological inputs [9,10]. Contemporary intelligent algorithms, particularly machine learning, and deep learning approaches have substantially enhanced the precision and efficiency of neural signal interpretation [11–13]. This technological maturation manifests in two primary BCI categories: non-invasive systems, particularly electroencephalography (EEG)-based interfaces, have exhibited considerable potential in regard to device control and communication facilitation [14,15], while invasive BCIs have achieved unprecedented breakthroughs in high-speed text and speech communication through precise motor cortex neural activity interpretation [6,8,16]. These advancements have facilitated a paradigm shift from fundamental research toward clinical applications, particularly in regard to rehabilitation support and assistive technologies for individuals with neuromuscular disorders [7,17].

Despite extensive research in this domain, existing reviews have predominantly focused on isolated aspects, namely neural encoding mechanisms [18,19], specific language decoding paradigms [20], clinical applications [7,17], or specific BCI modalities [21], while lacking a comprehensive evolutionary perspective that integrates cognitive mechanisms with technological implementations. While these focused reviews have provided valuable insights in regard to the respective areas, a broader integrative perspective would enhance our understanding of the crucial synergistic relationships between cognitive mechanisms and technological implementations, thereby advancing both our theoretical understanding and the potential for practical innovations in regard to BCI-based language decoding systems. To address this gap, we propose the Interpretation–Communication–Interaction (ICI) architecture, a novel three-stage evolutionary perspective for BCI-based language decoding (Figure 1). This architecture incorporates signal interpretation (basic stage), dynamic communication (advanced stage), and intelligent interaction (innovative stage), synthesizing cognitive neuroscience with computational methodologies. The subsequent sections examine the neurocognitive foundations of language decoding, analyze technological evolution through the ICI architecture lens, and explore future challenges and developmental trajectories in regard to BCI-based language decoding.

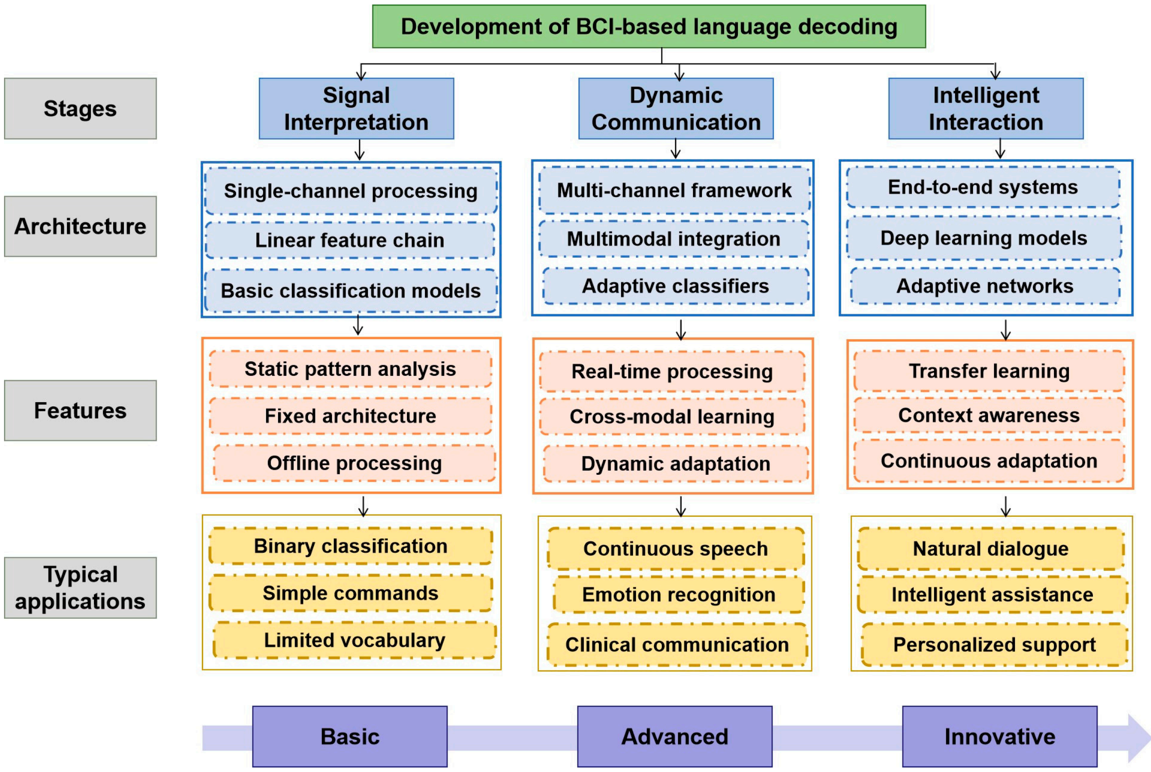


Figure 1. The Interpretation–Communication–Interaction (ICI) architecture, showing the three-stage evolution of BCI-based language decoding, from basic signal interpretation through advanced dynamic communication to innovative intelligent interaction, with corresponding architectures, features, and applications at each stage.

2. Neurocognitive Foundation of Language Decoding

The neurocognitive basis of BCI-based language decoding integrates three fundamental domains: cognitive mechanisms, a neural basis, and computational approaches. These domains cover cognitive and memory processes, brain network organization, and signal analysis methods, establishing the essential foundations for BCI language decoding.

2.1. Cognitive Mechanisms

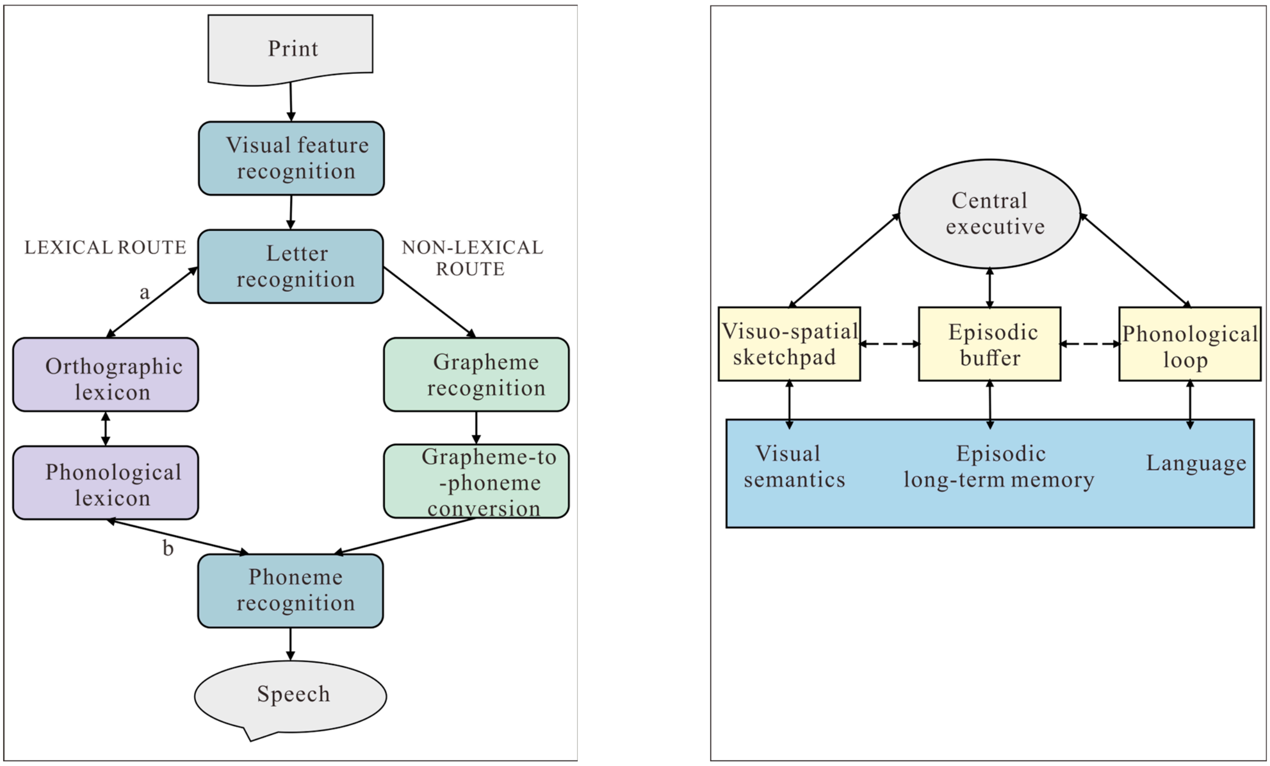
From a cognitive perspective, language decoding can be understood through three fundamental theoretical frameworks: information processing models, working memory architectures, and attentional control systems [22,23]. These interconnected cognitive models provide a systematic foundation for understanding how the brain processes, maintains, and selectively manages linguistic information, thereby informing BCI applications.

Information processing models have established crucial frameworks for understanding the cognitive mechanisms underlying language decoding, particularly in neural signal interpretation [24,25]. The dual-route framework, emerging from Marshall and Newcombe's seminal work on acquired dyslexia [26], suggests that reading processes operate through two distinct pathways. Based on their observations of patients with surface and deep dyslexia, they proposed that reading could proceed either through direct visual recognition or through phonological recoding. This theoretical framework evolved substantially with Coltheart et al.'s Dual-Route Cascaded (DRC) model in 2001, which elucidates visual word recognition and oral reading through two specialized pathways: a lexical route for processing familiar words and a non-lexical route for phonological conversion [27]. In their comprehensive review of reading models and dyslexia, Pritchard et al. provided a clear visualization of the DRC architecture that has been widely adopted in the field [28].

As illustrated in Figure 2a, the DRC model proposes two parallel cognitive pathways for reading processes: the lexical–semantic route and the grapheme–phoneme conversion route. These pathways operate simultaneously, yet independently, with each route contributing distinct computational outputs to stimulus articulation. The DRC model's computational implementation has significantly contributed to understanding reading disorders, such as dyslexia [29,30], particularly in explaining the dissociation between surface and phonological dyslexia patterns. Recent investigations by Divjak suggest that although the dual-route architecture remains valid, the traditional distinction between lexical and grammatical processing requires substantial reconsideration [31]. Building upon these processing frameworks, Friederici's temporal model illuminates the sequential nature of linguistic computation in neural systems [32], delineating three distinct temporal phases: rapid syntactic parsing (100–300 ms post-stimulus), semantic integration (300–500 ms), and final syntactic integration (500–1000 ms). Each phase operates within specific temporal windows and engages distinct neural networks, providing critical insights into language processing dynamics. Extending these neurobiological insights, computational advances in multilingual models like mT5 have demonstrated how dual-route frameworks can be effectively adapted for linguistic decoding in low-resource languages, broadening their applications in natural language processing [33]. These evolving perspectives on information processing have advanced our understanding of language decoding mechanisms. However, a comprehensive account of language processing must also consider how working memory systems support and regulate these cognitive operations.

The Working Memory Model, introduced by Baddeley and Hitch in 1974, represents another fundamental framework comprising interconnected components that orchestrate linguistic information processing. In their original model, they proposed three main components: the central executive as an attentional control system, aided by two subsidiary storage systems, namely the phonological loop for verbal–acoustic information and the visuo-spatial sketchpad for visual material [34]. Baddeley later refined this model in 2010 by adding a fourth component called the episodic buffer (Figure 2b) [35]. In this refined model, the central executive functions as the primary supervisory system, coordinating these subsystems and managing cognitive load during complex language tasks, while the phonological loop, consisting of a phonological store and an articulatory rehearsal mechanism, has been particularly well-studied in regard to language acquisition and processing. Recent investigations by Schwering et al. demonstrate that verbal working memory emerges through active engagement in language processes [36], emphasizing its dynamic and adaptive nature. Their findings suggest that working memory capacity is not merely a static container, but rather a flexible system that adapts to linguistic demands through experience-dependent plasticity. While these studies focus primarily on verbal processing, the integration of artificial subjectivity frameworks into models of human cognition, as proposed by Munir and Qazi, suggests innovative pathways to simulate and extend human-like memory functions in regard to cognitive systems, offering insights for both biological and AI applications [37]. Extending beyond theoretical models to empirical investigations, research by Marian et al. has revealed significant interactions between visuo-spatial and phonological processing, particularly in bilingual contexts [38]. Taken together, these findings underscore the integrated nature of sensory processing within working memory systems and their role in supporting complex linguistic tasks.

Attention mechanisms constitute the third fundamental neural substrate that modulates information processing and cognitive resource allocation in language processing. Corbetta et al. have identified two primary attention mechanisms: goal-directed attention operating through top-down cognitive control and stimulus-driven attention functioning through bottom-up processing [39]. These mechanisms rely on distinct but interacting neural networks that enable adaptive responses to varying linguistic demands. The dorsal attention network, including the frontal eye fields and intraparietal sulcus, supports goal-directed attention, while the ventral attention network, comprising the temporoparietal junction and ventral frontal cortex, mediates stimulus-driven attention [40,41]. The temporal dynamics of these attentional mechanisms have been extensively investigated through combined EEG and pupillometric analyses, where Lim et al. have demonstrated that attentional continuity exhibits significant modulations during speaker transitions, revealing specific temporal windows of heightened susceptibility to disruption [42]. These findings provide crucial insights for understanding how attention systems dynamically adapt to support language processing in naturalistic contexts.



(**a**) (**b**)

Figure 2. (**a**) The DRC model of reading processing (adapted from [28]), showing lexical (vocabulary-based) and non-lexical (letter-to-sound conversion) pathways from print to speech output. (**b**) The multicomponent Working Memory Model (adapted from [35]), illustrating interactions between the central executive system and its specialized subsystems (visuo-spatial, episodic, and phonological components), with long-term memory processes.

The integration of these three theoretical frameworks provides a comprehensive foundation for understanding language decoding mechanisms, from initial signal processing to higher-order cognitive operations. This integrated perspective not only advances our theoretical understanding, but also informs the development of BCIs and clinical interventions.

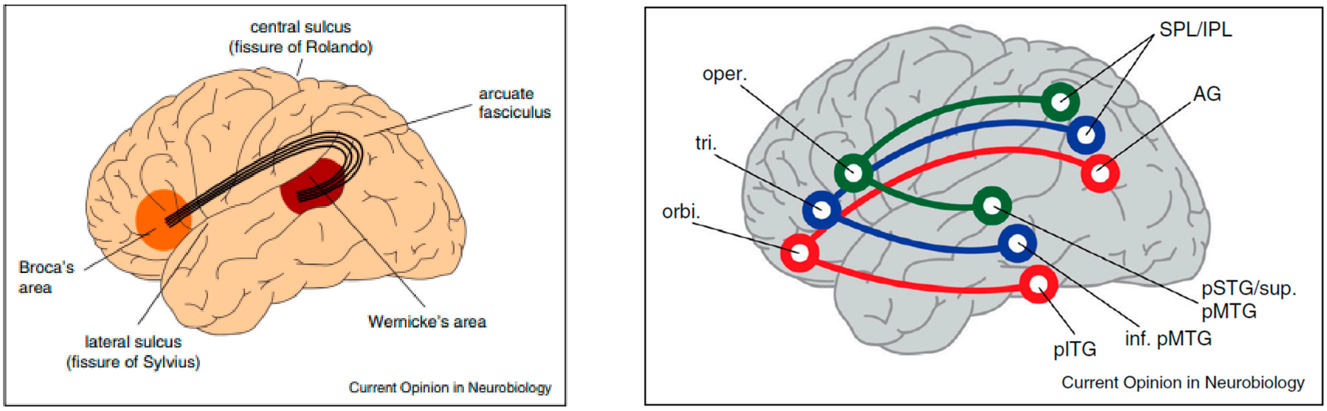
2.2. Neural Basis

The neural basis of language processing is fundamental for BCI-based language decoding. Key brain regions involved in language processing, their network connectivity patterns, and the spatiotemporal dynamics of neural activities during linguistic tasks provide essential insights for developing effective decoding strategies in regard to BCI applications. Understanding these neural mechanisms is crucial for BCI systems, as they inform the selection of optimal recording sites, guide the design of decoding algorithms, and enable more accurate interpretation of neural signals during language tasks.

The neural architecture of language processing comprises a sophisticated network of interconnected regions. For BCI-based language decoding, identifying and targeting these specific brain regions is essential for electrode placement and signal acquisition. Two primary areas, Broca's and Wernicke's areas, form the cornerstone of language function. Broca's area, situated in the left inferior frontal gyrus (Brodmann areas 44 and 45), orchestrates language production, grammatical processing, and articulatory planning [43]. Damage to this region results in expressive aphasia, characterized by impaired speech production, while preserving comprehension capabilities [44]. Wernicke's area, located in the posterior section of the superior temporal gyrus (Brodmann area 22), facilitates language comprehension and semantic processing, with lesions leading to receptive aphasia, marked by fluent but semantically impaired speech [45]. These critical language-processing regions and their functional connectivity patterns are illustrated in Figure 3a. While this classical Wernicke–Lichtheim–Geschwind model has dominated the field for over a century, recent evidence suggests that the distribution of labor between these regions is more complex than originally proposed, namely lesions in Broca's region impair not only language production but also comprehension, while Wernicke's region lesions affect both comprehension and production [46,47]. Nevertheless, the distinct functional roles of these regions provide valuable targets for BCI electrode arrays and inform the development of specialized decoding algorithms for different aspects of language processing. The prefrontal cortex, particularly the dorsolateral and ventrolateral regions, mediates higher-order linguistic functions, including executive control, working memory, and linguistic planning [48]. The temporal auditory regions, specifically Heschl's gyrus and the surrounding areas, process spectrotemporal features of speech signals and phonological representations [49]. The parietal lobe, especially the angular and supramarginal gyri, integrates multimodal linguistic information and facilitates semantic processing [50]. Recent neuroimaging studies have significantly advanced our understanding of language network organization. Through resting-state functional Magnetic Resonance Imaging (fMRI) analysis, Xiang et al. identified distinct topographical patterns of functional connectivity within the left inferior frontal, inferior parietal, and temporal regions [51]. Subsequently, Hagoort demonstrated that these connectivity patterns align with three core components of language processing in the perisylvian cortex: phonology, syntax, and semantics [52]. As illustrated in Figure 3b, this distributed network exhibits robust connections based on information type processing, providing crucial insights for BCI system design, particularly regarding the necessity of multi-site recording and integrated decoding approaches to capture comprehensive language-related neural activity.

Recent investigations have elucidated the neural network connectivity underlying language processing. Understanding these connectivity patterns is crucial for BCI systems to accurately decode intended speech from distributed neural signals and to develop more sophisticated network-based decoding algorithms. Evolutionary studies by Balezeau et al. have revealed homologous pathways across primates, establishing an ancestral prototype for the arcuate fasciculus and documenting its evolutionary transformation in supporting human language capabilities [53]. While the Wernicke–Lichtheim model has traditionally dominated our understanding of language's neuroanatomical foundation, contemporary neurolinguistic research has evolved toward a dual-stream language processing framework. Notably, Hickok et al. proposed a comprehensive functional neuroanatomical model that delineates bilateral temporal lobe pathways for speech perception, emphasizing the interface between auditory and motor systems [54]. This model was later expanded to encompass speech production and working memory mechanisms [55]. The framework's dual-stream architecture provides a theoretical basis for designing BCI systems that can simultaneously decode both speech production intentions and comprehension processes.

The dorsal stream, connecting temporal and frontal regions via the arcuate fasciculus, facilitates speech production and phonological processing, while the ventral stream, coursing through the extreme capsule fiber system, supports comprehension and semantic processing. Advanced diffusion tensor imaging techniques enabled Weiller et al. to identify supramodal systems underlying language processing [56], while Kümmerer et al. validated the dual-stream organization through systematic lesion-behavior mapping in stroke patients [57]. Hagoort delved into the dynamic network perspective of language, accentuating the interaction that occurs between core language regions and other networks, such as attention and a Theory of Mind, which is essential for realizing the full spectrum of language functionality [52]. Collectively, these studies underscore the complexity of the neural networks involved in language, suggesting an integrated, yet specialized, system across species.



(**a**) (**b**)

Figure 3. (**a**) The classical Wernicke–Lichtheim–Geschwind language network model (adapted from [52]), showing the core components of language processing including Broca's area (orange), Wernicke's area (red), and their connection via the arcuate fasciculus; (**b**) distributed connectivity pattern in the Perisylvian language network (adapted from [52]), illustrating information-specific pathways (green: syntactic, blue: semantic, red: phonological) between Broca's subregions and posterior language areas. SPL/IPL: superior/inferior parietal lobule; AG: angular gyrus; pSTG: posterior superior temporal gyrus; pMTG: posterior middle temporal gyrus; pITG: posterior inferior temporal gyrus.

The spatiotemporal dynamics of language processing exhibit sophisticated patterns of neural oscillations and information flow. These temporal patterns provide critical features for BCI systems to decode language intentions in real time and to distinguish between different linguistic processes. Giraud and Poeppel demonstrated how delta (1–3 Hz), theta (4–8 Hz), and gamma (30–80 Hz) band oscillations hierarchically parse incoming speech signals into meaningful units, with different frequency bands corresponding to distinct linguistic features, such as prosodic, syllabic, and phonemic information [58]. This frequency-specific organization offers BCI systems multiple channels for parallel decoding of different linguistic features. Bernstein et al.'s event-related potential (ERP) studies revealed simultaneous activation patterns across multiple regions during audiovisual speech processing [59]. The temporal precision of these activation patterns provides crucial timing information for BCI systems to synchronize decoding operations with natural language processing rhythms. Computational modeling by Bai et al. provided evidence of time-based binding mechanisms in structural encoding [60]. Furthermore, Leonard et al.'s neuroimaging studies of bilingual processing demonstrated differential activation patterns between first and second languages [61]. These findings inform the development of adaptive BCI systems that are capable of accommodating individual differences in language processing and handling multiple languages. The complex temporal organization of neural activity in language processing highlights the importance of considering both spatial and temporal dimensions in developing BCI-based decoding strategies, ultimately guiding the design of more sophisticated and effective neural decoding algorithms for language BCIs.

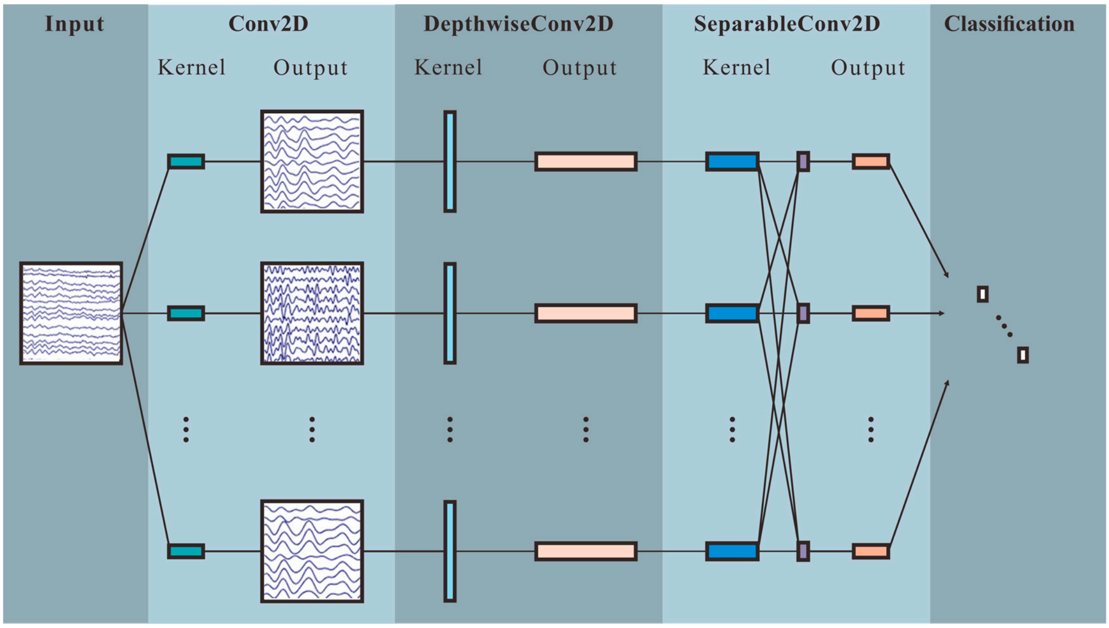
2.3. Computational Approaches

Computational approaches in regard to BCI-based language decoding comprise three fundamental components: feature extraction, classification algorithms, and performance evaluation. These methodological frameworks systematically transform raw neural signals into interpretable linguistic outputs through multi-stage processing pipelines. The process integrates temporal, spectral, and spatial analyses, followed by both conventional and deep learning-based classification methods, with performance quantified through comprehensive evaluation metrics. This systematic approach ensures robust signal processing and reliable decoding outcomes in real-world BCI applications.

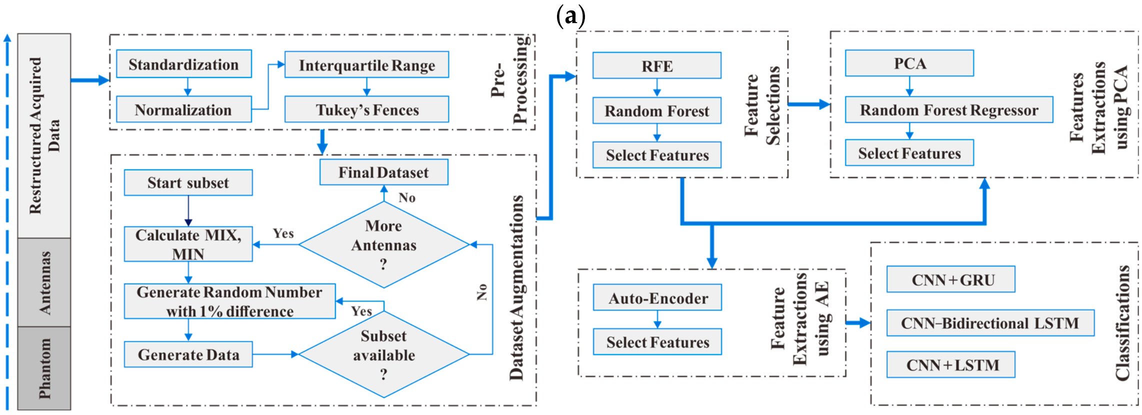
In the context of language decoding from neural signals, robust feature extraction methodologies are instrumental in ensuring reliable classification outcomes. These methods are particularly crucial for capturing language-related neural patterns in EEG signals, as different linguistic processes generate distinct spatiotemporal patterns. Spatial filtering techniques, particularly the common spatial pattern (CSP) algorithm, have demonstrated significant efficacy in enhancing single-trial EEG analysis. Ramoser et al. established that CSP effectively optimizes spatial filters to differentiate mental states, substantially improving motor imagery classification accuracy through adaptive spatial filter optimization [62]. This advancement has been particularly valuable for decoding intended speech and language components from neural signals. The evolution of feature extraction has been further accelerated by deep learning architecture. Notably, Lawhern et al. introduced EEGNet, a compact convolutional neural network (CNN) specifically engineered for EEG signal processing, which simultaneously extracts temporal and spectral features, while maintaining computational efficiency [63]. EEGNet implements a hierarchical signal transformation pathway that systematically processes neural signals from acquisition to interpretation. The architecture's information transmission process begins with raw EEG input signals and progresses through specialized convolutional layers, each serving a distinct purpose in regard to signal recognition and transformation. This systematic approach ensures comprehensive neural pattern capture, while maintaining signal fidelity throughout the processing pipeline. As illustrated in Figure 4a, EEGNet's architecture implements sequential processing stages: temporal convolution for frequency filtering, depthwise convolution for frequency-specific spatial filtering, and separable convolution for optimal feature map integration. This architecture has proven especially effective in capturing the complex temporal dynamics of language-related neural activities. Furthermore, integrated time–frequency analysis has emerged as a crucial approach for capturing EEG's multidimensional characteristics. In the context of language decoding, this is particularly important as different linguistic features (phonemes, words, and sentences) manifest across various temporal scales. Xie et al. proposed a dual-feature extraction network that effectively leverages both temporal and frequency information, establishing a new paradigm in EEG signal processing through innovative feature fusion mechanisms [64].

For the successful translation of neural signals into linguistic content, the evolution of classification algorithms has become paramount in BCI applications requiring precise neural signal interpretation. These algorithms are specifically designed to map extracted neural features to corresponding linguistic elements, such as phonemes, words, or semantic concepts. As depicted in Figure 4b, the classification pipeline encompasses a comprehensive sequence of processing stages, beginning with essential preprocessing steps, including data standardization, normalization, and outlier handling through the use of the interquartile range and Tukey's fences. The pipeline then incorporates a systematic dataset augmentation approach, which implements strategic data generation when additional training samples are needed. Following this, multiple parallel pathways for feature selection and extraction are employed, utilizing recursive feature elimination (RFE), random forest-based selection, principal component analysis (PCA), and auto-encoder-based feature extraction. The final stage involves classification using advanced deep learning architectures, specifically CNN + gated recurrent unit (GRU), CNN–bidirectional long short-term memory (LSTM), and CNN + LSTM for optimal decision making. While traditional machine learning approaches, such as support vector machine (SVM) and k-nearest neighbors (k-NN), have been fundamental, due to their interpretability and computational efficiency, they often encounter limitations in regard to complex, high-dimensional EEG signals [65]. To address these challenges, the neural signal classification pipeline implements a comprehensive approach to signal transformation and interpretation. This systematic process incorporates multiple stages of data preprocessing and feature engineering, utilizing both conventional and advanced machine learning techniques. The pipeline's design emphasizes robust signal processing through parallel pathways for feature selection and extraction, enabling the effective capturing of complex neural patterns, while maintaining computational efficiency. Furthermore, Du et al. developed an adaptive deep metric learning loss function optimized for class imbalance scenarios, enhancing both intra-class diversity and inter-class distillation through dynamic weight adjustment [66]. This methodology has demonstrated particular efficacy in EEG applications, where signal variance is substantial, and has shown promise in regard to diverse applications, including Alzheimer's detection [67] and cancer cell identification [68]. Similarly, Pawuś and Paszkiel explored the integration of EEG signals in the development of novel classification algorithms, tailored for real-time robotic control. Their pilot study demonstrated the utility of movement imagery-based EEG signal processing for enhancing classification accuracy, suggesting potential parallels in mapping neural signals to linguistic content for BCI-based language decoding [69]. In the context of BCI-based language decoding, hybrid classification strategies, combining traditional and deep learning approaches, have emerged as effective solutions, particularly in scenarios with limited labeled data, leveraging the strengths of both paradigms, while mitigating their respective limitations.

To ensure the reliability and accuracy of decoded linguistic content, performance evaluation in regard to EEG-based BCIs necessitates careful consideration of multiple metrics, due to the inherent complexity of neural signals. These metrics are specifically adapted to assess the accuracy of language decoding, including word recognition rates, semantic similarity measures, and linguistic coherence scores. While accuracy serves as a fundamental measure of classification performance, Luque et al. emphasized the importance of complementary metrics, such as precision, recall, and F1-score, particularly in addressing imbalanced datasets and ensuring comprehensive performance assessment [70]. In the context of language decoding, these metrics help evaluate both the accuracy of individual word recognition and the coherence of decoded sentences. These metrics provide a more comprehensive assessment of model performance by incorporating both true positives and false positives, enabling detailed analysis of classification behavior across different signal conditions. For language-specific BCI applications, temporal and spatial resolution metrics are equally crucial in evaluating EEG classification systems, and determining their ability to capture rapid neural dynamics and precise spatial distributions of brain activity. Blankertz et al. emphasized the importance of stability, especially for BCI systems used in real-world applications, where consistent performance across trials is key to user acceptance and practical deployment [71]. This stability is particularly critical for maintaining reliable communication through BCI-based language systems. Stability indicators, including inter-session reliability and cross-subject generalization performance, further enhance the robustness assessment of BCI systems through the systematic evaluation of temporal consistency and user adaptability. These comprehensive evaluation frameworks enable a systematic comparison across different methodologies, while ensuring the reproducibility and clinical applicability of BCI-based language decoding systems, ultimately facilitating the translation of research advances into practical applications.



(**a**)



(**b**)

Figure 4. (**a**) Sequential processing stages of EEGNet (adapted from [63]), showing temporal filtering (Conv2D), frequency-specific spatial filtering (DepthwiseConv2D), and feature map integration (SeparableConv2D), followed by classification. (**b**) An integrated deep learning pipeline for neural signal classification (adapted from [67]), demonstrating the sequential stages of data augmentation, feature selection, feature extraction, and classification.

3. Three-Stage Development of BCI-Based Language Decoding

The evolution of BCI-based language decoding exhibits a clear three-stage developmental trajectory. The basic stage focuses on signal interpretation, where neural signals are processed through fundamental computational architectures and static pattern recognition algorithms. This initial stage established essential methodologies for neural signal analysis and classification. The advanced stage marked significant progress in BCI systems, characterized by dynamic communication frameworks that integrate multimodal fusion techniques and real-time processing, enabling more sophisticated applications, such as continuous speech decoding. The innovative stage, driven by emerging AI technologies, introduces comprehensive platforms that synergize biological and AI. To conceptualize this evolutionary progression, we propose the ICI architecture (Figure 1). This section examines the core architecture, distinctive features, and principal applications that characterize each developmental stage, demonstrating how BCI-based language decoding has evolved from basic signal processing to sophisticated neural–artificial interfaces.

A comprehensive analysis of this evolutionary trajectory necessitates an examination of the fundamental signal transformation process that underlies all BCI-based language decoding systems. Table 1 presents an overview of the brain signal transformation process, illustrating the technological sophistication in BCI-based language decoding systems, particularly those utilizing EEG and electrocorticography (ECoG) signals. This process encompasses five interconnected stages that form the foundation for dynamic communication platforms. Beginning with neural signal acquisition through EEG/ECoG sensors, the pipeline progresses through sophisticated preprocessing and feature extraction phases, leveraging advanced signal processing techniques, such as independent component analysis (ICA) and wavelet transform. The subsequent decoding stage employs state-of-the-art machine learning algorithms and neural networks to interpret these processed signals, culminating in the output generation stage, where language models and speech synthesizers transform neural patterns into meaningful linguistic outputs. This comprehensive signal transformation framework underpins the enhanced real-time processing capabilities and multimodal fusion approaches that characterize advanced BCI systems, enabling more sophisticated and responsive communication platforms. The following subsections analyze how each developmental stage has advanced and optimized these fundamental transformation processes.

Table 1. Brain signal transformation process in regard to EEG/ECoG-based BCI language decoding. Wav2vec: wave-to-vector; CSP: common spatial pattern; ICA: independent component analysis.

Stage	Process	Key Components	Methods/Techniques
Signal acquisition	Raw neural data collection	EEG/ECoG sensors Recording devices	Multichannel recording High sampling rate acquisition
Preprocessing	Signal cleaning and enhancement	Artifact removal Noise reduction Signal filtering	ICA Band-pass filtering CSP
Feature extraction	Signal characteristics analysis	Temporal features Spectral features Spatial features	Time–frequency analysis Wavelet transform Deep learning (e.g., EEGNet)
Decoding	Neural pattern recognition	Classification Pattern matching Language modeling	Machine learning algorithms Deep neural networks wav2vec 2.0
Output generation	Language production	Text generation Speech synthesis Feedback mechanisms	Language models Speech synthesizers Real-time adaptation

3.1. Stage 1: Signal Interpretation, Basic Stage

The ICI architecture identifies signal interpretation as the basic stage of BCI-based language decoding (Figure 1), characterized by fundamental approaches focusing on single-channel processing, linear feature extraction methods, and basic classification models. These basic approaches are essential for extracting meaningful linguistic information from brain signals, as language processing involves complex temporal and spatial patterns across multiple brain regions. This initial phase employed basic pattern recognition techniques with static architectures and offline analysis paradigms, primarily achieving binary classification tasks and simple command recognition within controlled laboratory environments [72,73]. While limited in terms of complexity, these foundational methods established critical proof-of-concept tests for decoding language-related neural patterns. The applications, though limited in scope, established essential methodological frameworks for subsequent developments in neural signal processing and interpretation, particularly in the context of real-time brain signal analysis and feature extraction methodologies.

Building upon this foundational framework, the signal processing architecture in regard to EEG and other brain signal decoding tasks includes single-channel processing, linear feature extraction, traditional classification techniques, and offline analysis paradigms. These components are particularly crucial for detecting and analyzing the temporal dynamics of speech-related neural activities. Single-channel processing facilitates focused signal acquisition from individual sensors, which is particularly valuable in region-specific analysis, as demonstrated by Dmochowski et al. in their investigation of attention markers associated with emotionally laden engagement, through the use of precise temporal resolutions of EEG recordings [74]. Their work utilized inter-subject correlation analysis to identify neural synchronization patterns across multiple participants, establishing a robust methodology for analyzing shared neural responses to naturalistic stimuli. Linear dimensionality reduction methods, notably PCA and ICA, serve to extract critical features, while minimizing noise interference in high-dimensional neural data [75,76]. These techniques have been proven to be particularly effective in isolating task-relevant neural signatures from background activity and artifacts. For classification algorithm selection, Kotsiantis provided a comprehensive comparison framework, evaluating 13 key performance dimensions across major algorithms, including decision trees, neural networks, naïve bayes, k-NN, SVM, and rule learners [77]. As shown in Table 2, their analysis revealed that while SVMs excel in regard to general accuracy and handling irrelevant attributes, they require longer training time. Neural networks demonstrate strengths in processing highly interdependent attributes and incremental learning, but are less effective when dealing with missing values. These insights provide valuable guidance for choosing appropriate classifiers in regard to brain signal decoding tasks. The offline analysis paradigm, exemplified in Pei et al.'s ECoG studies, enables comprehensive examination of phoneme patterns in both spoken and imagined speech [78], establishing a methodological foundation for neural signal interpretation through detailed post hoc analysis of temporal and spatial patterns. Their research specifically highlighted the importance of high-gamma band activity (70–150 Hz) in encoding phonemic information.

Table 2. Comparative analysis of machine learning algorithms (adapted from [77]); the stars (★, ★★, ★★★, ★★★★) indicate performance levels from minimal to optimal.

	Decision Trees	**Neural Networks**	**Naïve Bayes**	**k-NN**	**SVM**	**Rule Learners**
Accuracy in general	★★	★★★	★	★★	★★★★	★★
Speed of learning with respect to number of attributes and the number of instances	★★★	★	★★★★	★★★★	★	★★
Speed of classification	★★★★	★★★★	★★★★	★	★★★★	★★★★
Tolerance to missing values	★★★	★	★★★★	★	★★	★★
Tolerance to irrelevant attributes	★★★	★	★★	★★	★★★★	★★

Table 2. *Cont.*

	Decision Trees	Neural Networks	Naïve Bayes	k-NN	SVM	Rule Learners
Tolerance to redundant attributes	★★	★★	★	★★	★★★	★★
Tolerance to highly interdependent attributes (e.g., parity problems)	★★	★★★	★	★	★★★	★★
Dealing with discrete/binary/continuous attributes	★★★★	★★★ (not discrete)	★★★ (not continuous)	★★★ (not directly discrete)	★★ (not discrete)	★★★ (not directly continuous)
Tolerance to noise	★★	★★	★★★	★	★★	★
Dealing with danger of overfitting	★★	★	★★★	★★★	★★	★★
Attempts in regard to incremental learning	★★	★★★	★★★★	★★★★	★★	★
Explanation ability/transparency of knowledge/classifications	★★★★	★	★★★★	★★	★	★★★★
Model parameter handling	★★★	★	★★★★	★★★	★	★★★

In accordance with these processing methodologies, foundational pattern recognition in EEG-based BCI research predominantly employs static classification techniques for non-dynamic datasets. This approach is particularly effective for decoding discrete linguistic elements (such as phonemes and words) from continuous brain signals, as well as corresponding cognitive and motor actions, as documented in Abiri et al.'s comprehensive review of EEG-based BCI paradigms and their implementation in controlled environments [79]. Their analysis encompassed various preprocessing techniques, including spatial filtering methods, such as CSP and Laplacian filtering, which enhance the signal-to-noise ratio and spatial resolution. The implementation of single-task learning protocols and fixed model architectures, while limiting adaptability, ensures computational efficiency and reproducibility in controlled laboratory settings, as highlighted by Lotte et al.'s systematic analysis of classification algorithms in BCI applications [80]. Their work specifically addressed the challenges of non-stationarity in EEG signals and proposed adaptive classification strategies to maintain the level of performance over extended usage periods. Fundamental statistical analyses of signal features, as demonstrated by Li et al. in silent reading and language processing tasks [81], provide essential insights into data distributions and patterns, establishing a robust framework for preliminary BCI applications through rigorous statistical validation methods. Their research employed time–frequency analysis techniques, including short-time Fourier transform (STFT) and wavelet decomposition, to characterize the spectral dynamics of language-related neural activity.

Extending beyond these methodological foundations, preliminary applications in regard to BCI-based language decoding primarily focus on binary classification tasks within constrained vocabulary domains. These early implementations laid the groundwork for more sophisticated language decoding systems, by demonstrating the feasibility of extracting linguistic information from neural signals. The widely implemented BCI2000 system exemplifies this approach through its application in regard to standardized two-choice tasks, demonstrating the feasibility of reliable neural signal classification in controlled settings [82]. This platform incorporates real-time signal processing modules, feature extraction algorithms, and classification methods within a unified framework, enabling the systematic evaluation of different BCI paradigms. Contemporary developments, including EEGNet, have enhanced decoding efficiency within these constrained parameters through optimized neural network architectures, specifically designed for EEG signal processing [63]. The architecture incorporates depthwise and separable convolutions, significantly reducing the number of trainable parameters, while maintaining high classification accuracy across various BCI paradigms. The practical implementation of these foundational approaches has been further validated through several clinical studies focusing on patients with severe motor disabilities. Naito et al. developed a near-infrared spectroscopy (NIRS)-based communication system for amyotrophic lateral sclerosis (ALS) patients in a totally locked-in state (TLS), achieving an accuracy of approximately 80% across 40 subjects through single-channel processing and basic discriminant analysis [83]. Building upon this approach, Gallegos-Ayala et al. successfully implemented a similar NIRS-based system for a completely locked-in ALS patient, utilizing single-channel processing and SVM classification, with positive and negative predictive values of 80.9% and 72.9%, respectively [84]. More recently, Rutten et al. validated these foundational methods through fMRI studies, demonstrating that even single-feature linear decoding approaches can effectively reconstruct speech-related neural patterns, with an accuracy of approximately 80% [85]. Beyond these clinical applications, the potential of neural signal processing has been further demonstrated in developmental research. The successful application of convolutional neural networks (CNNs) in decoding speech-related information from infant EEG signals demonstrates the potential in regard to fundamental language recognition capabilities, particularly in detecting and classifying basic linguistic features [86]. These implementations underscore the critical importance of controlled experimental conditions in ensuring the reliability and reproducibility of results, particularly given the inherent limitations of first-generation BCI systems, where performance optimization requires careful consideration of signal acquisition protocols, artifact reduction strategies, and the development of robust feature extraction pipelines that can handle the inherent variability in neural signals across different subjects and recording sessions.

Despite the promising potential of BCI-based language decoding, first-generation systems faced four major technical limitations that significantly hindered their practical applications in regard to neural signal decoding and interpretation. First, and foremost, these limitations included signal quality dependence, where system performance was heavily influenced by EEG signal quality [87], which significantly impacted the accuracy of linguistic feature extraction and interpretation. Beyond the signal quality issue, the limited generalization ability across users and tasks posed substantial challenges for cross-subject language decoding applications. This challenge was particularly pronounced as traditional methods lacked effective transfer learning techniques, making it difficult to adapt models to new users or tasks [88]. The third major challenge concerned high temporal delay-impeded real-time applications. This delay proved especially problematic for natural language communication scenarios, where immediate feedback and fluent interaction are essential [89]. Finally, and equally significant, are the restricted interaction capabilities hindering system performance in complex environments. This limitation particularly affected the development of practical language-based BCI applications, as natural communication typically occurs in dynamic, multimodal contexts [90]. These constraints emphasized the necessity for advancements in signal processing, generalization techniques, and real-time interaction mechanisms in subsequent BCI system developments, specifically targeting the unique challenges of neural-based language decoding and communication.

3.2. Stage 2: Dynamic Communication, Advanced Stage

The advanced stage of BCI-based language decoding demonstrates significant technological maturation through enhanced real-time processing architectures and multimodal fusion frameworks (Figure 1). These improvements enable the development of sophisticated dynamic communication platforms, leading to expanded applications and second-generation breakthroughs in regard to system performance and clinical implementation.

Building upon fundamental BCI signal acquisition and processing capabilities, enhanced real-time processing frameworks emerged as fundamental components for advanced system performance, marking a transition from traditional offline analysis to dynamic platforms. These advancements are particularly crucial for language decoding tasks, where real-time processing of complex linguistic neural patterns requires sophisticated computational architectures. The implemented frameworks incorporate multichannel parallel processing architectures, enabling simultaneous analysis of distributed neural signals across cortical regions to capture comprehensive brain activity patterns [91]. The implementation of dynamic feature extraction mechanisms, particularly advanced time–frequency analysis techniques, such as wavelet transform and adaptive spectral analysis, facilitates the identification of temporal variations in neural signals and accommodates the non-stationary nature of brain activity [92]. Furthermore, the integration of adaptive classification algorithms, notably the adaptive k-NN and dynamic SVM, addresses individual variability and signal fluctuations, thereby optimizing classification performance in dynamic conditions [93]. Building upon these algorithmic innovations, the incorporation of online learning mechanisms enables continuous system adaptation through real-time data-driven model updates, significantly enhancing the system's interactive capabilities and reducing the need for frequent calibration procedures.

Along with the evolution of signal processing techniques, the development of multimodal fusion frameworks marked a crucial advancement in BCI system architecture, representing a shift from single-modality approaches to integrated multi-source processing systems. This integration is especially vital for language decoding, as speech and language processing involve distributed neural networks that generate signals across multiple modalities, from motor cortex activity to semantic processing areas. These frameworks systematically integrate complementary information from multiple neural recording modalities, such as combined EEG–functional near-infrared spectroscopy (fNIRS) systems, substantially improving the performance of these frameworks across various cognitive and motor tasks, through the exploitation of both hemodynamic and electrophysiological signals [94]. To enhance the integration efficiency of the system, the implementation of cross-modal feature learning techniques, particularly deep learning architectures, including multimodal autoencoders and attention mechanisms, facilitates the extraction of shared representations from heterogeneous data sources, while preserving modality-specific characteristics. Notable contributions include the emotion recognition system proposed by Jiang et al., which employs dynamic weight adjustment across modalities to optimize the predictive accuracy and maintain system robustness in conditions with varying signal quality [95]. Moreover, collaborative enhancement mechanisms have been successfully implemented in clinical applications, such as in regard to Mild Cognitive Impairment (MCI) identification, where temporal and spatial features are integrated through adaptive dynamic functional connectivity analysis [96], demonstrating the practical utility of multimodal approaches in diagnostic settings.

Beyond these technical advances in signal processing and fusion, the expansion of BCI applications has led to unprecedented capabilities in neural signal interpretation and communication, extending beyond basic command-and-control paradigms to sophisticated language processing tasks. This expansion has been particularly transformative in the domain of language decoding, where advances in neural signal interpretation have enabled the direct decoding of attempted speech and linguistic intent from brain signals. The practical implementation of these advances has been demonstrated through several clinical studies, showcasing significant progress in continuous speech decoding and sentence-level comprehension. A comprehensive clinical evaluation by McCane et al. assessed 25 individuals with advanced ALS, using a P300-based BCI system [97]. Their findings revealed that 72% of participants achieved accuracy rates above 70%, sufficient for basic communication, despite severe physical disabilities. Notably, performance was not correlated with disability severity, with many participants with an ALSF functional rating scale revised score below 5 achieving high accuracy rates, demonstrating the system's potential in regard to severely impaired patients. Another clinical implementation was reported by Herff et al., who developed a "brain-to-text" system that successfully decoded continuous speech from intracranial recordings [98]. Testing on seven patients with epilepsy showed word error rates as low as 25% for vocabulary sizes of 100 words. The system demonstrated robust performance, even with limited training data, as little as 25 min of speech was sufficient for accurate decoding. Most recently, Anumanchipalli et al. achieved a breakthrough in speech synthesis, using neural signals in a clinical setting [99]. Their system, tested on five participants undergoing intracranial monitoring, successfully synthesized intelligible speech by leveraging advanced neural network architectures to decode intended vocal tract movements from brain signals. Notably, the system could synthesize speech, even during silent miming, achieving correlation scores of 0.69 with actual speech patterns, demonstrating its potential for helping patients who cannot physically vocalize. These clinical implementations have collectively demonstrated robust performance across diverse patient populations, reduced training requirements, and successful operation in challenging clinical environments, establishing the practical viability of BCI-based language decoding systems. In parallel with these foundational clinical successes in regard to linguistic advances, the integration of emotion recognition systems has enhanced the sophistication of user interactions, exemplified by Li et al.'s implementation of functional connectivity networks combined with local activation features for EEG-based emotion recognition, enabling more naturalistic and context-aware communication [100]. Complementary to these developments, Lai and Tseng demonstrated the effectiveness of deep learning models in analyzing user sentiments, semantic features, and preferences in terms of product recommendations, underscoring the potential of combining sentiment and semantic analysis with neural decoding systems to further personalize BCI communication platforms and align them with user-specific contexts [101]. Enriching the communication paradigm, the incorporation of contextual information processing, as demonstrated by Chen et al.'s implementation of sentence-level topics in the context of neural machine translation, has further advanced the system's comprehension capabilities by considering broader linguistic and semantic relationships [102]. These developments collectively contribute to more intuitive and expressive BCI-based communication systems.

As BCI technology continues to mature, modern systems have achieved significant improvements across multiple domains, including enhanced reliability, reduced calibration requirements, and improved user experience. These improvements have been particularly impactful in regard to language decoding applications, where enhanced system reliability and reduced latency are crucial for natural speech production and comprehension. Notable progress has been achieved in real-time response optimization, as evidenced by Anumanchipalli et al.'s demonstration of direct neural decoding for spoken sentence reconstruction, which has substantially reduced processing latency, while maintaining high decoding accuracy [99]. This innovation has expanded the therapeutic possibilities for individuals with speech impairments, establishing new protocols for rehabilitation and assistive communication. System robustness has been substantially enhanced through advanced signal processing methodologies and improved model generalization. Wu et al.'s comprehensive analysis of transfer learning applications in regard to EEG-based BCIs revealed significant progress in addressing inter-subject variability and environmental adaptability [103]. These developments have led to more reliable and consistent performance across diverse user populations and operational conditions. The enhancement of human–computer interaction paradigms has been achieved through sophisticated interface designs and adaptive feedback mechanisms. The integration of deep neural architectures and multimodal data streams has facilitated more intuitive and responsive user experiences, particularly in regard to real-time applications. These advancements have directly contributed to expanded clinical applications, notably in regard to therapeutic interventions and diagnostic procedures. The implementation of emotion recognition systems in clinical settings has demonstrated particular efficacy in regard to mental health monitoring and neurological assessment protocols, establishing new standards for psychological evaluation and treatment monitoring.

3.3. Stage 3: Intelligent Interaction, Innovation Stage

The intelligent interaction stage represents the culmination of the ICI architecture's evolution in regard to BCI-based language decoding, characterized by sophisticated neural decoding systems that achieve efficient brain-to-text conversion through integrated architecture and adaptive learning capabilities (Figure 1). By integrating advanced deep learning frameworks with end-to-end neural signal processing, this stage establishes a synergistic network ecosystem that operates through transfer learning and continuous adaptation mechanisms. Technological advancement manifests through three key dimensions: intelligent architecture integration, sophisticated learning strategies, and next-generation capabilities. Together, these elements drive the transformation from conventional signal-processing approaches toward autonomous, context-aware systems that excel in regard to dynamic adaptation and personalized interactions.

To realize this vision of intelligent interaction and establish the foundation for advanced neural decoding, the architectural framework of modern BCI-based language decoding systems has evolved through the systematic integration of complementary technological innovations. The implementation of contemporary deep learning frameworks, particularly PyTorch, has fundamentally transformed neural decoder development by introducing dynamic computational graphs and imperative programming paradigms, establishing a robust foundation for advanced decoding algorithms [104]. This sophisticated technical infrastructure has catalyzed the emergence of multi-task collaborative learning in regard to EEG-based neural decoding, where shared representations across linguistic tasks significantly enhance decoder robustness and the generalization capacity, as extensively documented by Craik et al. [105]. Building upon these technological foundations, the integration of advanced linguistic frameworks has marked another significant milestone in BCI-based language decoding systems. Modern architectures now incorporate sophisticated linguistic processing layers that seamlessly integrate phonological, syntactic, and semantic analysis, enabling a more nuanced understanding of neural signals associated with language processing. This advancement is particularly evident in regard to the recent breakthrough by Chen et al., who developed a novel end-to-end framework that leverages interpretable acoustic parameters to bridge neural activity and speech production [106]. Their approach combines an ECoG decoder with a differentiable speech synthesizer, demonstrating how careful parameter design at multiple linguistic levels, from fundamental frequency to formant structures, can enable accurate and natural speech reconstruction across a large cohort of 48 participants. While Chen et al.'s work demonstrates the potential of invasive ECoG recordings, parallel breakthroughs have been achieved in regard to non-invasive BCI systems, as shown through developments in end-to-end systems, exemplified by Défossez et al.'s innovative approach to direct speech perception decoding using non-invasive brain recordings [14]. As shown in Figure 5, they proposed an end-to-end architecture that combines wave-to-vector (wav2vec) 2.0, a pre-trained speech module, with a specialized brain module. The system utilizes contrastive language–image pre-training (CLIP) loss to align brain activity representations with speech features, enabling zero-shot decoding of previously unseen speech segments, a significant advancement over traditional classification approaches. Défossez et al.'s work demonstrated the potential of using pre-trained speech models for non-invasive brain decoding. While their approach focused on speech perception, subsequent research has explored different directions to advance BCI systems, including Tang et al.'s work on semantic reconstruction using non-invasive brain recordings [107], which demonstrated that natural language decoders trained on fMRI data can capture semantic meaning across different cognitive tasks, including perceived speech, imagined speech, and even visual narratives. Their approach leverages pre-trained language models and encoding–decoding frameworks to enable more accurate interpretation of neural signals and enhanced semantic modeling. In parallel, Dong et al. introduced another innovative methodology by addressing neural signal non-stationarity through dynamic online learning mechanisms [18], thereby establishing a comprehensive framework that optimizes performance across various operational conditions, while maintaining system stability and reliability.

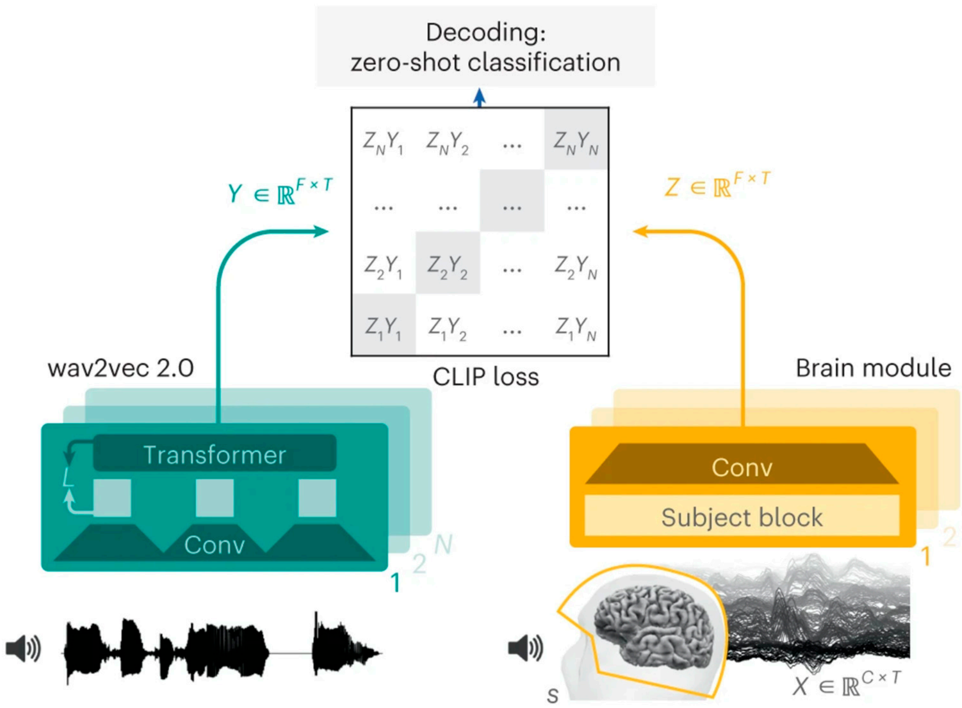


Figure 5. End-to-end architecture for decoding speech from brain signals using wav2vec 2.0 and contrastive learning (adapted from [14]); wav2vec: wave-to-vector; CLIP: contrastive language–image pre-training; Conv: convolution.

Advances in learning strategies have emerged as another crucial direction in BCI-based language decoding, with novel training approaches and optimization techniques specifically addressing the inherent challenges of neural signal variability and inter-subject differences. Transfer learning methodologies have revolutionized the field by enabling efficient model adaptation across subjects with minimal calibration requirements, substantially reducing the system initialization time, while improving cross-subject generalization [103].

Concurrent developments in regard to continuous learning mechanisms have addressed the fundamental stability–plasticity dilemma in neural networks, enabling progressive knowledge accumulation, while preserving established functionalities [108]. Significant advances in regard to rapid adaptation protocols have emerged through various innovative approaches to neural speech decoding. These developments include hybrid architectures incorporating recurrent neural networks and transformer models, as demonstrated in several studies, including Chen et al.'s work [106], which represents one of the implementations combining deep learning with speech synthesis. In parallel, Wang et al. established new paradigms in continuous learning through their innovative implementation of physical neuromorphic computing [109], representing a significant advancement in continuous knowledge integration, while maintaining system stability.

Leveraging these advanced architectural foundations and sophisticated learning strategies, the integration of next-generation functionalities has propelled BCI-based language decoding systems toward increasingly sophisticated and personalized implementations. The convergence of these technological capabilities has enabled the development of more intuitive and responsive communication interfaces that can adapt to individual user needs and environmental contexts. Natural language understanding capabilities have become fundamental to modern BCI systems, with Cooney et al.'s neurolinguistic approaches demonstrating significant improvements in direct speech decoding accuracy [110]. This advancement synergizes with enhanced context awareness, as emphasized in Müller-Putz et al.'s research on hybrid BCI systems [111], establishing more robust and reliable language interpretation protocols that can adapt to varying communication scenarios. The incorporation of intelligent decision support mechanisms, as investigated by Schirner et al., has optimized network structures in regard to bio-inspired computing, significantly enhancing system adaptability across diverse communication contexts, while maintaining processing efficiency [112]. Notable progress in clinical applications has been demonstrated through several recent studies. Angrick et al. implemented a chronically implanted BCI system that achieved 80% intelligibility in synthesizing speech from neural signals involving an ALS patient [113]. Their three-stage approach using recurrent neural networks enabled reliable decoding of self-paced speech commands from ECoG signals across motor, premotor, and somatosensory cortices. The system's stability was validated over a 6-week period, demonstrating long-term viability for clinical applications [113]. Chen et al. further advanced clinical implementation by developing a neural speech decoding framework that was tested across 48 participants, achieving correlation coefficients of 0.62–0.92 between decoded and ground-truth speech [106]. Their framework incorporated both causal and non-causal architectures, with the causal models showing comparable performance (PCC = 0.797), critical for real-time clinical applications. These clinical validations were complemented by Shoaran et al.'s development of intelligent neural interfaces integrating on-implant signal processing and AI, demonstrating successful implementation of closed-loop symptom tracking and responsive stimulation in regard to various neurological conditions [114]. These studies collectively provide robust empirical evidence for the practical deployment of individualized language decoding solutions in clinical settings.

The convergence of these capabilities, including natural language understanding, context awareness, intelligent decision support, and personalized customization, establishes a comprehensive framework that advances BCI-based language decoding toward more intuitive, adaptive, and user-centric solutions, significantly enhancing the communication capabilities of individuals with neurological conditions, while paving the way for future innovations in regard to human–computer interactions.

3.4. Evolution Pattern Analysis and Comparison

The evolution patterns of BCI-based language decoding demonstrate systematic progression across technical, performance, and application dimensions, reflecting the transformative advancement in neural communication technologies. Through comprehensive analysis of evolutionary trajectories, this investigation reveals the intricate interplay between technological innovation, performance optimization, and practical implementation in regard to neural decoding systems.

The technical evolution trajectory exhibits multidimensional progression, characterized by increasingly sophisticated methodological approaches. Rather than discrete jumps between stages, the field demonstrates continuous advancement through transitional phases. The progression from basic to advanced stages was marked by a gradual evolution in both signal processing and computational frameworks, exemplified by Schalk et al.'s implementation of parallel processing and sophisticated feature selection through stepwise linear discriminant analysis (SWLDA) [115], and Leuthardt et al.'s expansion from single to multi-phoneme classification, with advanced 8×8 electrode configurations [116]. In regard to signal processing, the field has witnessed a decisive transition from rudimentary single-modality methods through multimodal integration to advanced intelligent fusion systems, enabling substantially more robust language decoding performance [117]. This methodological advancement parallels the evolution in algorithmic architectures, progressing from conventional machine learning paradigms through hybrid computational models to sophisticated deep learning frameworks, optimized for neural signal interpretation [118]. The transition to innovative stages has been characterized by architectural advancement, demonstrated by Makin et al.'s introduction of encoder–decoder frameworks [119] and Tang et al.'s development of semantic reconstruction approaches [107]. The synergistic development of signal processing and algorithmic innovations has catalyzed the transformation of practical applications, evolving from fundamental offline analysis to real-time processing capabilities and, ultimately, to intelligent interactive systems. Contemporary biomimetic computer-to-brain communication systems exemplify this progression, achieving unprecedented naturalistic sensory feedback, through intelligent neural signal processing and interpretation [120]. System characteristics have evolved correspondingly, transcending the initial confined laboratory applications to become increasingly universal and, ultimately, personalized, with adaptive neural speech decoding frameworks demonstrating remarkable capabilities in regard to individual user customization and optimization [106].

Performance metrics have demonstrated distinct developmental trajectories throughout multiple critical parameters across the three ICI stages (Table 3), establishing new benchmarks in regard to neural communication efficiency. These improvements reflect gradual transitions, rather than sudden breakthroughs. McCane et al.'s standardization of recording protocols with 8- and 16-channel montages exemplified a crucial intermediate step in signal processing maturation [97]. Subsequently, Duraivel et al.'s implementation of high-resolution micro-ECoG recordings demonstrated the progressive enhancement of spatial resolution and signal quality [121]. Fundamentally, despite increasing task complexity across the different stages, the decoding performance has maintained its stability. As shown in Table 3, the signal interpretation stage achieved a consistent level of performance (80–100%) in basic binary tasks, while dynamic communication expanded to complex speech tasks and, in doing so, maintained considerable accuracy (25–98% accuracy). Notably, intelligent interaction demonstrated enhanced stability (45.6–91.8%), even as it advanced to sophisticated natural communication tasks. Technologies, such as high-density micro-electrocorticographic (μECoG) recordings, exemplify these improvements, in regard to enhanced spatial resolution and signal quality [121]. In parallel, temporal processing capabilities have advanced in regard to neural decoding systems. As shown in Table 3, latencies improved from 3 to 26 s in signal interpretation's basic tasks to 2.64–10.23 ms in dynamic communication's speech decoding. Despite handling more complex natural language tasks, intelligent interaction maintains efficient performance at 50–235 ms. Recent implementations using spiking neural networks (SNNs) have demonstrated the potential for further optimization, achieving sub-10 ms processing, with high computational efficiency [122]. Correspondingly, system stability has improved substantially in regard to neural decoding systems. As shown in Table 3, performance consistency evolved from single-session stability in signal interpretation to cross-session reliability in dynamic communication, with accuracy maintained at 96.8% across the sessions. The intelligent interaction stage further achieved consistent performance (PCC > 0.8) across 25 subjects. In particular, long-term stability has been demonstrated in intracortical BCIs, which maintain robust recording capabilities for up to 3 years in non-human primates and 5 years in human subjects subject to controlled conditions [18]. Likewise, user adaptation requirements have substantially evolved. As shown in Table 3, the training duration was reduced from 5 to 7 weeks in regard to the signal interpretation stage to 20–107 min in regard to the dynamic communication stage. The intelligent interaction stage shows varied adaptation periods from 15 min to 6 weeks, reflecting different task complexities. Recent automatic calibration methods have achieved efficient parameter updates at 2–5 s intervals [109].

Table 3. Chronological comparison of BCI systems' performance metrics across three development stages of ICI architecture. N/A: not available; ECoG: electrocorticography; PCC: Pearson correlation coefficient; ALS: amyotrophic lateral sclerosis; ★ indicates transitional work between development stages.

Stage	Source	Task	Performance Metrics			
			Accuracy	Latency	Stability	User Adaptation Time
Signal Interpretation (Basic)	[123]	P300-based matrix speller	95%	26 s	All four subjects achieved 95% accuracy	2–3 sessions
	[124]	Cursor control based on mu rhythm	80–95%	3 s	N/A	15–20 sessions (5–7 weeks)
	[125]	Visual–motor decision task	80–100%	291–676 ms	N/A	N/A
Dynamic Communication (Advanced)	★ [97]	Visual P300-based matrix speller	12–92%	N/A	N/A	60–90 min
	[98]	Brain-to-text decoding was recorded	50–75%	N/A	N/A	20–48 min
	[126]	Real-time classification of auditory sentences	90–98%	2.64–10.23 ms	N/A	6.5–11 min
	[99]	Speech synthesis from ECoG signals	25–90%	N/A	Models effective based on novel test data	27–107 min
Intelligent Interaction (Innovative)	★ [121]	Speech production task with non-words	20.9–56.7%	N/A	Up to 15 min recording duration	15 min
	[127]	Speech decoding from neural signals	76.2–90.9%	N/A	High correlation between nearby days	177.3–184.7 min
	[8]	Speech decoding and avatar control	45.6–91.8%	200 ms	96.8% accuracy maintained across sessions	14 days

Table 3. *Cont.*

Stage	Source	Task	Performance Metrics			
			Accuracy	Latency	Stability	User Adaptation Time
Intelligent Interaction (Innovative)	[106]	Speech decoding from ECoG signals	71.2–80.6%	50 ms	25 subjects maintained a PCC > 0.8	N/A
	[113]	Speech synthesis from ECoG signals in ALS patient	80–93.4%	235 ms	Effective after 5.5 months based on 6-week training	6 weeks

In parallel with these technical and performance improvements, the scope of application has expanded considerably, validating and exceeding the ICI architecture's evolutionary predictions. BCI systems have evolved from basic communication aids to sophisticated comprehensive solutions integrating advanced speech decoding and synthesis capabilities, facilitating increasingly natural communication modalities [7]. The user base has diversified significantly beyond the initial constraints, as evidenced by extensive clinical trials of implantable BCIs, extending beyond specialized research subjects to encompass diverse patient populations with varying communication needs and capabilities [128]. The functional complexity has progressed systematically from basic neural signal decoding to sophisticated mechanistic models incorporating multiple neural representations and processing pathways [117], while the practical value has increased substantially through enhanced reliability and effectiveness in real-world applications across diverse environmental conditions [13].

The systematic analysis of these evolutionary trajectories reveals several significant emerging trends that warrant careful consideration in terms of future development efforts. First, the accelerating convergence of technical capabilities and performance metrics suggests an imminent breakthrough in practical application feasibility, potentially leading to widespread clinical implementation. Second, the increasing emphasis on personalization and adaptability indicates a fundamental shift towards user-centric design paradigms, necessitating more sophisticated adaptive algorithms and interface solutions. Third, the expanding scope of application demonstrates the technology's potential beyond traditional medical applications to broader communication contexts, suggesting possible paradigm shifts in human–machine interactions. Fourth, the integration of advanced machine learning techniques with neurophysiological insights indicates a trend toward more biologically informed computational models. These trends collectively suggest that BCI-based language decoding is approaching a critical transition point from experimental technology to practical communication solutions, with profound implications for both clinical applications and future human–machine interaction paradigms.

4. Key Challenges and Development Trends

As BCI-based language decoding advances, several critical challenges demand attention across technical, clinical, and social dimensions. Current barriers include fundamental technical limitations in regard to signal quality and processing efficiency, rigorous requirements for clinical safety and efficacy validation, and complex ethical considerations for real-world deployment. These challenges, coupled with broader social implications, significantly influence the field's technological evolution and prospects.

4.1. Technical Bottlenecks

The development of BCI systems, particularly for language decoding applications, encounters several critical technical bottlenecks, including signal quality enhancement, real-time optimization, interference resistance, and system integration.

Signal quality represents a fundamental challenge in regard to BCI systems, especially when decoding complex neural patterns associated with speech. Neural recordings are inherently susceptible to various artifacts, including physiological activities and environmental interference, which significantly impact decoding accuracy [129,130]. Recent advances in signal processing have introduced sophisticated methods for artifact identification and removal in regard to multifractal signals, marking substantial progress in terms of EEG signal quality enhancement [131]. This advancement is particularly crucial for maintaining reliable decoding performance in practical applications where signal contamination is prevalent.

The real-time processing capability presents another critical challenge, particularly in regard to speech and language decoding applications, where minimal latency is essential for natural communication. While high-resolution neural recordings, such as intracranial EEG and ECoG, have demonstrated improved decoding accuracy [121], the challenge lies in processing these complex signals efficiently. The requirement for simultaneous processing of multiple data channels, while maintaining both speed and accuracy, remains a significant technical hurdle [132,133].

Interference resistance constitutes a crucial aspect of BCI system development. In language decoding applications, the challenge is particularly acute as speech-related neural signals must be distinguished from concurrent cognitive processes and motor activities associated with natural communication. The ability to discriminate between relevant neural signals and various forms of interference, both external and physiological, is essential. Recent research has highlighted the potential of advanced signal processing and machine learning techniques in regard to non-invasive brain recordings for speech perception decoding, emphasizing their role in interference mitigation [14]. These developments are instrumental in ensuring robust system performance across diverse operational environments.

System integration represents the culmination of these technical challenges, requiring seamless coordination between hardware components and software algorithms. The complexity of language decoding demands an integrated approach where signal acquisition, processing, and interpretation work in concert to extract meaningful linguistic content from neural signals. Earlier research has analyzed the integration of language models with BCI classifiers, exemplifying the necessity of combining sophisticated neural signal processing with predictive modeling to enhance communication systems [134]. This foundational work continues to influence current approaches in the field. Addressing these technical bottlenecks necessitates continued advancement in regard to signal processing algorithms, hardware optimization, and machine learning models for neural signal interpretation. The progressive resolution of these challenges will facilitate the development of more efficient, real-time BCI systems, suitable for both clinical and non-clinical applications.

4.2. Clinical Translation

The clinical translation of BCI-based language decoding systems necessitates addressing multifaceted challenges to facilitate their integration into medical practice [128]. These challenges cover three primary dimensions, namely safety considerations, protocol standardization and validation procedures, and ethical frameworks, each requiring systematic attention for successful clinical implementation.

Safety evaluation represents a primary concern, particularly for invasive BCI implementations in regard to language decoding applications. Mitchell et al. conducted a comprehensive safety assessment of a fully implanted endovascular BCI in patients with severe paralysis, demonstrating promising safety outcomes [135]. This study, focusing on a device known as the Stentrode, underscores the critical importance of rigorous safety protocols, particularly in managing potential complications, such as infection risks and hardware reliability, during extended clinical deployment.

Protocol standardization and validation procedures constitute interconnected requirements for clinical translation. Clinical viability demands robust empirical evidence, obtained through systematic testing in authentic medical environments. Recent research in the field emphasizes the significance of standardized protocols for EEG-based BCI user training, highlighting how structured methodologies enhance the consistency and reliability of the results [136]. These protocols enable systematic optimization of BCI systems for individual patients, while facilitating meaningful cross-study outcome comparisons.

Ethical framework development represents the third critical dimension of clinical translation. A focused study has examined key ethical considerations surrounding BCI technologies, including privacy protection, patient autonomy, and informed consent procedures [137]. From a legal perspective, recent research by Brocal has further emphasized the critical need for robust data protection protocols and secure transmission mechanisms [138], as BCIs are increasingly vulnerable to security breaches that could compromise sensitive neural data. This legal dimension extends to the development of clear regulatory frameworks governing data handling and patient privacy rights. The social challenges in BCI deployment have been systematically examined by King et al. [139] and Awuah et al. [140], in particular, highlighting issues of patient autonomy and social equity. Their work reveals that obtaining meaningful informed consent requires careful consideration, especially for patients with varying cognitive capacity, while also emphasizing the potential impact on human identity and the risk of exacerbating social inequalities through differential access to BCI technology. The ethical implications, as analyzed by Livanis et al., underscore the complex interplay between technological advancement and social responsibility, particularly regarding emerging challenges, such as cognitive enhancement applications and the need for international standardization of ethical guidelines [141]. These works collectively contribute to frameworks guiding the responsible deployment of BCI technologies in clinical settings, emphasizing the importance of addressing legal protections, social implications, and ethical considerations as interconnected aspects of successful clinical translation.

The systematic resolution of these three key challenges, safety, protocol standardization and validation, and ethics, will significantly advance the clinical translation of BCI systems, ultimately expanding their therapeutic applications.

4.3. Future Prospects

Future prospects in regard to BCI-based language decoding cover multiple dimensions of advancement, including technological innovation, application expansion, and systematic implementation frameworks.

The field of BCI-based language decoding stands at a pivotal juncture in terms of its technological evolution, poised for significant advancement through emerging technologies and expanding applications. Recent innovations in neural signal processing have demonstrated remarkable progress, particularly through the integration of triboelectric sensors with deep learning algorithms, substantially enhancing decoding precision and reliability [142]. The incorporation of large pre-trained language models presents a promising direction for improving contextual comprehension and system adaptability in regard to language interpretation tasks [143].

The expansion of application domains represents another significant development trajectory. Advanced implementations, including lip-reading technologies and multimodal decoding interfaces, are establishing new paradigms in assistive technologies, human–computer interactions, and educational applications [142,144]. This diversification of applications underscores the versatility of language processing systems across multiple sectors, from clinical interventions to interactive entertainment platforms.

The transition from research protocols to commercial implementation necessitates scalable frameworks and standardized methodologies. Recent comprehensive surveys of BCI technologies emphasize the importance of establishing robust standardization protocols and scalable architectural designs to facilitate this transition [15]. Furthermore, the optimization of production processes and economic viability analysis remain crucial considerations for successful commercialization [128].

These technological trajectories and implementation strategies indicate a promising future for BCI-based language decoding systems, suggesting potential breakthroughs in human–machine communication and societal applications. The convergence of advanced signal processing, expanding use cases, and standardized deployment frameworks, positions this field in readiness to have a significant impact in regard to enhancing human communication capabilities.

5. Conclusions

The field of BCI-based language decoding has undergone a transformative evolution, progressing through three distinct stages: from fundamental signal interpretation to dynamic communication and, ultimately, toward intelligent interaction. Our analysis has identified crucial links between cognitive models of language processing, neural substrates, and computational approaches, demonstrating how this interdisciplinary integration has driven BCI advancements. This evolution is characterized by transitions from single-channel to multimodal signal processing, from traditional algorithms to deep learning architectures, and from basic functionalities to intelligent, personalized systems. Throughout this development, significant improvements have been achieved in regard to system accuracy, latency reduction, stability, and user adaptability, enabling its expansion from simple binary classification tasks to complex natural language understanding and context-aware interactions.

Looking forward, several critical directions emerge for advancing BCI-based language decoding systems. The complex nature of language processing necessitates enhanced collaboration between neuroscientists, computer scientists, linguists, and clinicians, particularly in developing personalized models that can adapt to individual users' neural patterns and linguistic preferences. As these systems become more sophisticated, addressing ethical concerns regarding privacy and data security becomes crucial, alongside the need to bridge the gap between laboratory research and clinical applications. Future developments should focus on the multimodal integration of neural and physiological signals, the incorporation of cutting-edge AI technologies, and conducting longitudinal studies to understand the long-term effects on neural plasticity. The continued evolution of this field holds tremendous promise for enhancing human–computer interactions and assisting individuals with communication disabilities, with success ultimately depending on the synergistic integration of neuroscience, AI, and clinical applications.

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References

1. Evans, N.; Levinson, S.C. The Myth of Language Universals: Language Diversity and Its Importance for Cognitive Science. *Behav. Brain Sci.* **2009**, *32*, 429–448. [CrossRef]
2. Friederici, A.D. The Brain Basis of Language Processing: From Structure to Function. *Physiol. Rev.* **2011**, *91*, 1357–1392. [CrossRef]
3. Anderson, A.J.; Kiela, D.; Binder, J.R.; Fernandino, L.; Humphries, C.J.; Conant, L.L.; Raizada, R.D.S.; Grimm, S.; Lalor, E.C. Deep Artificial Neural Networks Reveal a Distributed Cortical Network Encoding Propositional Sentence-Level Meaning. *J. Neurosci.* **2021**, *41*, 4100. [CrossRef] [PubMed]
4. Goldstein, A.; Zada, Z.; Buchnik, E.; Schain, M.; Price, A.; Aubrey, B.; Nastase, S.A.; Feder, A.; Emanuel, D.; Cohen, A.; et al. Shared Computational Principles for Language Processing in Humans and Deep Language Models. *Nat. Neurosci.* **2022**, *25*, 369–380. [CrossRef] [PubMed]
5. Schrimpf, M.; Blank, I.; Tuckute, G.; Kauf, C.; Hosseini, E.A.; Kanwisher, N.; Tenenbaum, J.; Fedorenko, E. The Neural Architecture of Language: Integrative Modeling Converges on Predictive Processing. *bioRxiv* **2021**. [CrossRef]
6. Willett, F.R.; Avansino, D.T.; Hochberg, L.R.; Henderson, J.M.; Shenoy, K.V. High-Performance Brain-to-Text Communication via Handwriting. *Nature* **2021**, *593*, 249–254. [CrossRef]
7. Luo, S.; Rabbani, Q.; Crone, N.E. Brain-Computer Interface: Applications to Speech Decoding and Synthesis to Augment Communication. *Neurotherapeutics* **2022**, *19*, 263–273. [CrossRef] [PubMed]
8. Metzger, S.L.; Littlejohn, K.T.; Silva, A.B.; Moses, D.A.; Seaton, M.P.; Wang, R.; Dougherty, M.E.; Liu, J.R.; Wu, P.; Berger, M.A.; et al. A High-Performance Neuroprosthesis for Speech Decoding and Avatar Control. *Nature* **2023**, *620*, 1037–1046. [CrossRef] [PubMed]
9. Lebedev, M.A.; Nicolelis, M.A.L. Brain-Machine Interfaces: From Basic Science to Neuroprostheses and Neurorehabilitation. *Physiol. Rev.* **2017**, *97*, 767–837. [CrossRef]
10. Tang, C.; Xu, Z.; Occhipinti, E.; Yi, W.; Xu, M.; Kumar, S.; Virk, G.S.; Gao, S.; Occhipinti, L.G. From Brain to Movement: Wearables-Based Motion Intention Prediction across the Human Nervous System. *Nano Energy* **2023**, *115*, 108712. [CrossRef]
11. Schirrmeister, R.T.; Springenberg, J.T.; Fiederer, L.D.J.; Glasstetter, M.; Eggensperger, K.; Tangermann, M.; Hutter, F.; Burgard, W.; Ball, T. Deep Learning with Convolutional Neural Networks for EEG Decoding and Visualization. *Hum. Brain Mapp.* **2017**, *38*, 5391–5420. [CrossRef] [PubMed]
12. Merk, T.; Peterson, V.; Köhler, R.; Haufe, S.; Richardson, R.M.; Neumann, W.-J. Machine Learning Based Brain Signal Decoding for Intelligent Adaptive Deep Brain Stimulation. *Exp. Neurol.* **2022**, *351*, 113993. [CrossRef] [PubMed]
13. Hu, H.; Wang, Z.; Zhao, X.; Li, R.; Li, A.; Si, Y.; Wang, J.; Zhou, T.; Xu, T. A Survey on Brain-Computer Interface-Inspired Communications: Opportunities and Challenges. *IEEE Commun. Surv. Tutor.* **2024**, *27*, 1. [CrossRef]
14. Défossez, A.; Caucheteux, C.; Rapin, J.; Kabeli, O.; King, J.-R. Decoding Speech Perception from Non-Invasive Brain Recordings. *Nat. Mach. Intell.* **2023**, *5*, 1097–1107. [CrossRef]
15. Edelman, B.J.; Zhang, S.; Schalk, G.; Brunner, P.; Müller-Putz, G.; Guan, C.; He, B. Non-Invasive Brain-Computer Interfaces: State of the Art and Trends. *IEEE Rev. Biomed. Eng.* **2024**, *17*, 1–25. [CrossRef] [PubMed]
16. Card, N.S.; Wairagkar, M.; Iacobacci, C.; Hou, X.; Singer-Clark, T.; Willett, F.R.; Kunz, E.M.; Fan, C.; Vahdati Nia, M.; Deo, D.R.; et al. An Accurate and Rapidly Calibrating Speech Neuroprosthesis. *N. Engl. J. Med.* **2024**, *391*, 609–618. [CrossRef]
17. Silva, A.B.; Littlejohn, K.T.; Liu, J.R.; Moses, D.A.; Chang, E.F. The Speech Neuroprosthesis. *Nat. Rev. Neurosci.* **2024**, *25*, 473–492. [CrossRef] [PubMed]
18. Dong, Y.; Wang, S.; Huang, Q.; Berg, R.W.; Li, G.; He, J. Neural Decoding for Intracortical Brain–Computer Interfaces. *Cyborg Bionic Syst.* **2024**, *4*, 0044. [CrossRef]
19. Frisby, S.L.; Halai, A.D.; Cox, C.R.; Lambon Ralph, M.A.; Rogers, T.T. Decoding Semantic Representations in Mind and Brain. *Trends Cogn. Sci.* **2023**, *27*, 258–281. [CrossRef] [PubMed]
20. Tang, J.; Chen, J.; Xu, X.; Liu, A.; Chen, X. Imagined Speech Reconstruction From Neural Signals—An Overview of Sources and Methods. *IEEE Trans. Instrum. Meas.* **2024**, *73*, 1–21. [CrossRef]
21. Alday, P.M. M/EEG Analysis of Naturalistic Stories: A Review from Speech to Language Processing. *Lang. Cogn. Neurosci.* **2019**, *34*, 457–473. [CrossRef]
22. Duncan, J. The Multiple-Demand (MD) System of the Primate Brain: Mental Programs for Intelligent Behaviour. *Trends Cogn. Sci.* **2010**, *14*, 172–179. [CrossRef] [PubMed]
23. Baddeley, A.D.; Eysenck, M.W.; Anderson, M.C. *Memory*; A Psychology Press Book; Taylor & Francis: Abingdon, UK, 2020; ISBN 9781138326071.
24. Frost, R. Towards a Universal Model of Reading. *Behav. Brain Sci.* **2012**, *35*, 263–279. [CrossRef]
25. Rastle, K. The Place of Morphology in Learning to Read in English. *Cortex* **2019**, *116*, 45–54. [CrossRef]
26. Marshall, J.C.; Newcombe, F. Patterns of Paralexia: A Psycholinguistic Approach. *J. Psycholinguist. Res.* **1973**, *2*, 175–199. [CrossRef]
27. Coltheart, M.; Rastle, K.; Perry, C.; Langdon, R.; Ziegler, J. DRC: A Dual Route Cascaded Model of Visual Word Recognition and Reading Aloud. *Psychol. Rev.* **2001**, *108*, 204–256. [CrossRef]

28. Pritchard, S.C.; Coltheart, M.; Palethorpe, S.; Castles, A. Nonword Reading: Comparing Dual-Route Cascaded and Connectionist Dual-Process Models with Human Data. *J. Exp. Psychol. Hum. Percept. Perform.* **2012**, *38*, 1268–1288. [CrossRef] [PubMed]
29. Coltheart, M. Dual Route and Connectionist Models of Reading: An Overview. *London Rev. Educ.* **2006**, *4*, 5–17. [CrossRef]
30. Castles, A.; Rastle, K.; Nation, K. Ending the Reading Wars: Reading Acquisition From Novice to Expert. *Psychol. Sci. Public Interest* **2018**, *19*, 5–51. [CrossRef] [PubMed]
31. Divjak, D.; Milin, P.; Medimorec, S.; Borowski, M. Behavioral Signatures of Memory Resources for Language: Looking beyond the Lexicon/Grammar Divide. *Cogn. Sci.* **2022**, *46*, e13206. [CrossRef]
32. Friederici, A.D. Towards a Neural Basis of Auditory Sentence Processing. *Trends Cogn. Sci.* **2002**, *6*, 78–84. [CrossRef] [PubMed]
33. Fuad, A.; Al-Yahya, M. AraConv: Developing an Arabic Task-Oriented Dialogue System Using Multi-Lingual Transformer Model MT5. *Appl. Sci.* **2022**, *12*, 1881. [CrossRef]
34. Baddeley, A.D.; Hitch, G. *Working Memory*; Bower, G.H., Ed.; Academic Press: Cambridge, MA, USA, 1974; Volume 8, pp. 47–89, ISBN 0079-7421.
35. Baddeley, A. Working Memory. *Curr. Biol.* **2010**, *20*, R136–R140. [CrossRef]
36. Schwering, S.C.; MacDonald, M.C. Verbal Working Memory as Emergent from Language Comprehension and Production. *Front. Hum. Neurosci.* **2020**, *14*, 68. [CrossRef] [PubMed]
37. Munir, A.; Qazi, W.M. Artificial Subjectivity: Personal Semantic Memory Model for Cognitive Agents. *Appl. Sci.* **2022**, *12*, 1903. [CrossRef]
38. Marian, V.; Hayakawa, S.; Schroeder, S.R. Memory after Visual Search: Overlapping Phonology, Shared Meaning, and Bilingual Experience Influence What We Remember. *Brain Lang.* **2021**, *222*, 105012. [CrossRef] [PubMed]
39. Corbetta, M.; Shulman, G.L. Control of Goal-Directed and Stimulus-Driven Attention in the Brain. *Nat. Rev. Neurosci.* **2002**, *3*, 201–215. [CrossRef]
40. Fox, M.D.; Corbetta, M.; Snyder, A.Z.; Vincent, J.L.; Raichle, M.E. Spontaneous Neuronal Activity Distinguishes Human Dorsal and Ventral Attention Systems. *Proc. Natl. Acad. Sci. USA* **2006**, *103*, 10046–10051. [CrossRef]
41. Alves, P.N.; Forkel, S.J.; Corbetta, M.; Thiebaut de Schotten, M. The Subcortical and Neurochemical Organization of the Ventral and Dorsal Attention Networks. *Commun. Biol.* **2022**, *5*, 1343. [CrossRef] [PubMed]
42. Lim, S.-J.; Carter, Y.D.; Njoroge, J.M.; Shinn-Cunningham, B.G.; Perrachione, T.K. Talker Discontinuity Disrupts Attention to Speech: Evidence from EEG and Pupillometry. *Brain Lang.* **2021**, *221*, 104996. [CrossRef]
43. Hagoort, P. On Broca, Brain, and Binding: A New Framework. *Trends Cogn. Sci.* **2005**, *9*, 416–423. [CrossRef]
44. Poeppel, D.; Hickok, G. Towards a New Functional Anatomy of Language. *Cognition* **2004**, *92*, 1–12. [CrossRef]
45. Wernicke, C. *The Symptom Complex of Aphasia BT—Proceedings of the Boston Colloquium for the Philosophy of Science 1966/1968*; Cohen, R.S., Wartofsky, M.W., Eds.; Springer: Dordrecht, The Netherlands, 1969; pp. 34–97, ISBN 978-94-010-3378-7.
46. Matchin, W.; Hickok, G. The Cortical Organization of Syntax. *Cereb. Cortex* **2020**, *30*, 1481–1498. [CrossRef]
47. Fedorenko, E.; Blank, I.A. Broca's Area Is Not a Natural Kind. *Trends Cogn. Sci.* **2020**, *24*, 270–284. [CrossRef] [PubMed]
48. Duncan, J.; Owen, A.M. Common Regions of the Human Frontal Lobe Recruited by Diverse Cognitive Demands. *Trends Neurosci.* **2000**, *23*, 475–483. [CrossRef]
49. Ding, N.; Melloni, L.; Zhang, H.; Tian, X.; Poeppel, D. Cortical Tracking of Hierarchical Linguistic Structures in Connected Speech. *Nat. Neurosci.* **2016**, *19*, 158–164. [CrossRef]
50. Teixeira, S.; Machado, S.; Velasques, B.; Sanfim, A.; Minc, D.; Peressutti, C.; Bittencourt, J.; Budde, H.; Cagy, M.; Anghinah, R.; et al. Integrative Parietal Cortex Processes: Neurological and Psychiatric Aspects. *J. Neurol. Sci.* **2014**, *338*, 12–22. [CrossRef] [PubMed]
51. Xiang, H.-D.; Fonteijn, H.M.; Norris, D.G.; Hagoort, P. Topographical Functional Connectivity Pattern in the Perisylvian Language Networks. *Cereb. Cortex* **2010**, *20*, 549–560. [CrossRef] [PubMed]
52. Hagoort, P. Nodes and Networks in the Neural Architecture for Language: Broca's Region and Beyond. *Curr. Opin. Neurobiol.* **2014**, *28*, 136–141. [CrossRef] [PubMed]
53. Balezeau, F.; Wilson, B.; Gallardo, G.; Dick, F.; Hopkins, W.; Anwander, A.; Friederici, A.D.; Griffiths, T.D.; Petkov, C.I. Primate Auditory Prototype in the Evolution of the Arcuate Fasciculus. *Nat. Neurosci.* **2020**, *23*, 611–614. [CrossRef] [PubMed]
54. Hickok, G.; Poeppel, D. Towards a Functional Neuroanatomy of Speech Perception. *Trends Cogn. Sci.* **2000**, *4*, 131–138. [CrossRef]
55. Hickok, G.; Poeppel, D. Dorsal and Ventral Streams: A Framework for Understanding Aspects of the Functional Anatomy of Language. *Cognition* **2004**, *92*, 67–99. [CrossRef]
56. Weiller, C.; Reisert, M.; Glauche, V.; Musso, M.; Rijntjes, M. The Dual-Loop Model for Combining External and Internal Worlds in Our Brain. *Neuroimage* **2022**, *263*, 119583. [CrossRef] [PubMed]
57. Kümmerer, D.; Hartwigsen, G.; Kellmeyer, P.; Glauche, V.; Mader, I.; Klöppel, S.; Suchan, J.; Karnath, H.-O.; Weiller, C.; Saur, D. Damage to Ventral and Dorsal Language Pathways in Acute Aphasia. *Brain* **2013**, *136*, 619–629. [CrossRef] [PubMed]
58. Giraud, A.-L.; Poeppel, D. Cortical Oscillations and Speech Processing: Emerging Computational Principles and Operations. *Nat. Neurosci.* **2012**, *15*, 511–517. [CrossRef]

59. Bernstein, L.E.; Auer, E.T.; Wagner, M.; Ponton, C.W. Spatiotemporal Dynamics of Audiovisual Speech Processing. *Neuroimage* **2008**, *39*, 423–435. [CrossRef]
60. Bai, F.; Meyer, A.S.; Martin, A.E. Neural Dynamics Differentially Encode Phrases and Sentences during Spoken Language Comprehension. *PLoS Biol.* **2022**, *20*, e3001713. [CrossRef] [PubMed]
61. Leonard, M.K.; Brown, T.T.; Travis, K.E.; Gharapetian, L.; Hagler, D.J.; Dale, A.M.; Elman, J.L.; Halgren, E. Spatiotemporal Dynamics of Bilingual Word Processing. *Neuroimage* **2010**, *49*, 3286–3294. [CrossRef]
62. Ramoser, H.; Muller-Gerking, J.; Pfurtscheller, G. Optimal Spatial Filtering of Single Trial EEG during Imagined Hand Movement. *IEEE Trans. Rehabil. Eng.* **2000**, *8*, 441–446. [CrossRef]
63. Lawhern, V.J.; Solon, A.J.; Waytowich, N.R.; Gordon, S.M.; Hung, C.P.; Lance, B.J. EEGNet: A Compact Convolutional Neural Network for EEG-Based Brain–Computer Interfaces. *J. Neural Eng.* **2018**, *15*, 56013. [CrossRef]
64. Xie, W.; Lei, J.; Fang, S.; Li, Y.; Jia, X.; Li, M. Dual Feature Extraction Network for Hyperspectral Image Analysis. *Pattern Recognit.* **2021**, *118*, 107992. [CrossRef]
65. Pfurtscheller, G.; Neuper, C. Motor Imagery and Direct Brain-Computer Communication. *Proc. IEEE* **2001**, *89*, 1123–1134. [CrossRef]
66. Du, J.; Zhang, X.; Liu, P.; Vong, C.-M.; Wang, T. An Adaptive Deep Metric Learning Loss Function for Class-Imbalance Learning via Intraclass Diversity and Interclass Distillation. *IEEE Trans. Neural Netw. Learn. Syst.* **2024**, *35*, 15372–15386. [CrossRef]
67. Farhatullah; Chen, X.; Zeng, D.; Ullah, R.; Nawaz, R.; Xu, J.; Arslan, T. A Deep Learning Approach for Non-Invasive Alzheimer's Monitoring Using Microwave Radar Data. *Neural Netw.* **2025**, *181*, 106778. [CrossRef]
68. Kandasamy, V.; Simic, V.; Bacanin, N.; Pamucar, D. Optimized Deep Learning Networks for Accurate Identification of Cancer Cells in Bone Marrow. *Neural Netw.* **2025**, *181*, 106822. [CrossRef]
69. Pawuś, D.; Paszkiel, S. The Application of Integration of EEG Signals for Authorial Classification Algorithms in Implementation for a Mobile Robot Control Using Movement Imagery—Pilot Study. *Appl. Sci.* **2022**, *12*, 2161. [CrossRef]
70. Luque, A.; Carrasco, A.; Martín, A.; de las Heras, A. The Impact of Class Imbalance in Classification Performance Metrics Based on the Binary Confusion Matrix. *Pattern Recognit.* **2019**, *91*, 216–231. [CrossRef]
71. Blankertz, B.; Lemm, S.; Treder, M.; Haufe, S.; Müller, K.-R. Single-Trial Analysis and Classification of ERP Components—A Tutorial. *Neuroimage* **2011**, *56*, 814–825. [CrossRef] [PubMed]
72. Birbaumer, N.; Ghanayim, N.; Hinterberger, T.; Iversen, I.; Kotchoubey, B.; Kübler, A.; Perelmouter, J.; Taub, E.; Flor, H. A Spelling Device for the Paralysed. *Nature* **1999**, *398*, 297–298. [CrossRef]
73. Leuthardt, E.C.; Schalk, G.; Wolpaw, J.R.; Ojemann, J.G.; Moran, D.W. A Brain–Computer Interface Using Electrocorticographic Signals in Humans*. *J. Neural Eng.* **2004**, *1*, 63. [CrossRef] [PubMed]
74. Dmochowski, J.P.; Sajda, P.; Dias, J.; Parra, L.C. Correlated Components of Ongoing EEG Point to Emotionally Laden Attention—A Possible Marker of Engagement? *Front. Hum. Neurosci.* **2012**, *6*, 112. [CrossRef] [PubMed]
75. Makeig, S.; Jung, T.-P.; Bell, A.J.; Ghahremani, D.; Sejnowski, T.J. Blind Separation of Auditory Event-Related Brain Responses into Independent Components. *Proc. Natl. Acad. Sci. USA* **1997**, *94*, 10979–10984. [CrossRef] [PubMed]
76. Delorme, A.; Makeig, S. EEGLAB: An Open Source Toolbox for Analysis of Single-Trial EEG Dynamics Including Independent Component Analysis. *J. Neurosci. Methods* **2004**, *134*, 9–21. [CrossRef] [PubMed]
77. Kotsiantis, S.B. Supervised Machine Learning: A Review of Classification Techniques. In *Emerging Artificial Intelligence Applications in Computer Engineering*; Maglogiannis, I., Ed.; IOS Press: Amsterdam, The Netherlands, 2007; Volume 160, pp. 3–24, ISBN 978-1-60750-270-8.
78. Pei, X.; Barbour, D.L.; Leuthardt, E.C.; Schalk, G. Decoding Vowels and Consonants in Spoken and Imagined Words Using Electrocorticographic Signals in Humans. *J. Neural Eng.* **2011**, *8*, 46028. [CrossRef] [PubMed]
79. Abiri, R.; Borhani, S.; Sellers, E.W.; Jiang, Y.; Zhao, X. A Comprehensive Review of EEG-Based Brain–Computer Interface Paradigms. *J. Neural Eng.* **2019**, *16*, 11001. [CrossRef] [PubMed]
80. Lotte, F.; Bougrain, L.; Cichocki, A.; Clerc, M.; Congedo, M.; Rakotomamonjy, A.; Yger, F. A Review of Classification Algorithms for EEG-Based Brain–Computer Interfaces: A 10 Year Update. *J. Neural Eng.* **2018**, *15*, 31005. [CrossRef]
81. Li, C.; Fang, G.; Liu, Y.; Liu, J.; Song, L. Decoding Silent Reading EEG Signals Using Adaptive Feature Graph Convolutional Network. *IEEE Signal Process. Lett.* **2024**, *31*, 1–5. [CrossRef]
82. Schalk, G.; McFarland, D.J.; Hinterberger, T.; Birbaumer, N.; Wolpaw, J.R. BCI2000: A General-Purpose Brain-Computer Interface (BCI) System. *IEEE Trans. Biomed. Eng.* **2004**, *51*, 1034–1043. [CrossRef]
83. Naito, M.; Michioka, Y.; Ozawa, K.; Ito, Y.I.; Kiguchi, M.; Kanazawa, T. A Communication Means for Totally Locked-in ALS Patients Based on Changes in Cerebral Blood Volume Measured with near-Infrared Light. *IEICE Trans. Inf. Syst.* **2007**, *90*, 1028–1037. [CrossRef]
84. Gallegos-Ayala, G.; Furdea, A.; Takano, K.; Ruf, C.A.; Flor, H.; Birbaumer, N. Brain Communication in a Completely Locked-in Patient Using Bedside near-Infrared Spectroscopy. *Neurology* **2014**, *82*, 1930–1932. [CrossRef]

85. Rutten, S.; Santoro, R.; Hervais-Adelman, A.; Formisano, E.; Golestani, N. Cortical Encoding of Speech Enhances Task-Relevant Acoustic Information. *Nat. Hum. Behav.* **2019**, *3*, 974–987. [CrossRef]
86. Keshavarzi, M.; Choisdealbha, Á.N.; Attaheri, A.; Rocha, S.; Brusini, P.; Gibbon, S.; Boutris, P.; Mead, N.; Olawole-Scott, H.; Ahmed, H.; et al. Decoding Speech Information from EEG Data with 4-, 7- and 11-Month-Old Infants: Using Convolutional Neural Network, Mutual Information-Based and Backward Linear Models. *J. Neurosci. Methods* **2024**, *403*, 110036. [CrossRef] [PubMed]
87. Lebedev, M.A.; Nicolelis, M.A.L. Brain–Machine Interfaces: Past, Present and Future. *Trends Neurosci.* **2006**, *29*, 536–546. [CrossRef]
88. He, H.; Wu, D. Transfer Learning for Brain–Computer Interfaces: A Euclidean Space Data Alignment Approach. *IEEE Trans. Biomed. Eng.* **2020**, *67*, 399–410. [CrossRef] [PubMed]
89. Chaudhary, U.; Birbaumer, N.; Ramos-Murguialday, A. Brain–Computer Interfaces for Communication and Rehabilitation. *Nat. Rev. Neurol.* **2016**, *12*, 513–525. [CrossRef] [PubMed]
90. Keutayeva, A.; Abibullaev, B. Data Constraints and Performance Optimization for Transformer-Based Models in EEG-Based Brain-Computer Interfaces: A Survey. *IEEE Access* **2024**, *12*, 62628–62647. [CrossRef]
91. Li, M.; Xu, H.; Liu, X.; Lu, S. Emotion Recognition from Multichannel EEG Signals Using K-Nearest Neighbor Classification. *Technol. Health Care* **2018**, *26*, 509–519. [CrossRef] [PubMed]
92. Hong, K.-S.; Khan, M.J.; Hong, M.J. Feature Extraction and Classification Methods for Hybrid FNIRS-EEG Brain-Computer Interfaces. *Front. Hum. Neurosci.* **2018**, *12*, 246. [CrossRef] [PubMed]
93. Mullick, S.S.; Datta, S.; Das, S. Adaptive Learning-Based k -Nearest Neighbor Classifiers With Resilience to Class Imbalance. *IEEE Trans. Neural Netw. Learn. Syst.* **2018**, *29*, 5713–5725. [CrossRef] [PubMed]
94. Sun, Z.; Huang, Z.; Duan, F.; Liu, Y. A Novel Multimodal Approach for Hybrid Brain–Computer Interface. *IEEE Access* **2020**, *8*, 89909–89918. [CrossRef]
95. Jiang, Y.; Li, W.; Hossain, M.S.; Chen, M.; Alelaiwi, A.; Al-Hammadi, M. A Snapshot Research and Implementation of Multimodal Information Fusion for Data-Driven Emotion Recognition. *Inf. Fusion* **2020**, *53*, 209–221. [CrossRef]
96. Li, Y.; Liu, J.; Tang, Z.; Lei, B. Deep Spatial-Temporal Feature Fusion From Adaptive Dynamic Functional Connectivity for MCI Identification. *IEEE Trans. Med. Imaging* **2020**, *39*, 2818–2830. [CrossRef]
97. McCane, L.M.; Sellers, E.W.; McFarland, D.J.; Mak, J.N.; Carmack, C.S.; Zeitlin, D.; Wolpaw, J.R.; Vaughan, T.M. Brain-Computer Interface (BCI) Evaluation in People with Amyotrophic Lateral Sclerosis. *Amyotroph. Lateral Scler. Front. Degener.* **2014**, *15*, 207–215. [CrossRef]
98. Herff, C.; Heger, D.; de Pesters, A.; Telaar, D.; Brunner, P.; Schalk, G.; Schultz, T. Brain-to-Text: Decoding Spoken Phrases from Phone Representations in the Brain. *Front. Neurosci.* **2015**, *9*, 1–11. [CrossRef]
99. Anumanchipalli, G.K.; Chartier, J.; Chang, E.F. Speech Synthesis from Neural Decoding of Spoken Sentences. *Nature* **2019**, *568*, 493–498. [CrossRef] [PubMed]
100. Li, P.; Liu, H.; Si, Y.; Li, C.; Li, F.; Zhu, X.; Huang, X.; Zeng, Y.; Yao, D.; Zhang, Y.; et al. EEG Based Emotion Recognition by Combining Functional Connectivity Network and Local Activations. *IEEE Trans. Biomed. Eng.* **2019**, *66*, 2869–2881. [CrossRef] [PubMed]
101. Lai, C.-H.; Tseng, K.-C. Applying Deep Learning Models to Analyze Users' Aspects, Sentiment, and Semantic Features for Product Recommendation. *Appl. Sci.* **2022**, *12*, 2118. [CrossRef]
102. Chen, K.; Wang, R.; Utiyama, M.; Sumita, E.; Zhao, T. Neural Machine Translation With Sentence-Level Topic Context. *IEEE/ACM Trans. Audio Speech Lang. Process.* **2019**, *27*, 1970–1984. [CrossRef]
103. Wu, D.; Xu, Y.; Lu, B.-L. Transfer Learning for EEG-Based Brain–Computer Interfaces: A Review of Progress Made Since 2016. *IEEE Trans. Cogn. Dev. Syst.* **2022**, *14*, 4–19. [CrossRef]
104. Paszke, A.; Gross, S.; Massa, F.; Lerer, A.; Bradbury, J.; Chanan, G.; Killeen, T.; Lin, Z.; Gimelshein, N.; Antiga, L.; et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. In *Advances in Neural Information Processing Systems 32*; Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., Garnett, R., Eds.; Curran Associates, Inc.: Red Hook, NY, USA, 2019; Volume 32, pp. 8024–8035.
105. Craik, A.; He, Y.; Contreras-Vidal, J.L. Deep Learning for Electroencephalogram (EEG) Classification Tasks: A Review. *J. Neural Eng.* **2019**, *16*, 31001. [CrossRef]
106. Chen, X.; Wang, R.; Khalilian-Gourtani, A.; Yu, L.; Dugan, P.; Friedman, D.; Doyle, W.; Devinsky, O.; Wang, Y.; Flinker, A. A Neural Speech Decoding Framework Leveraging Deep Learning and Speech Synthesis. *Nat. Mach. Intell.* **2024**, *6*, 467–480. [CrossRef]
107. Tang, J.; LeBel, A.; Jain, S.; Huth, A.G. Semantic Reconstruction of Continuous Language from Non-Invasive Brain Recordings. *Nat. Neurosci.* **2022**, *26*, 858–866.
108. Parisi, G.I.; Kemker, R.; Part, J.L.; Kanan, C.; Wermter, S. Continual Lifelong Learning with Neural Networks: A Review. *Neural Netw.* **2019**, *113*, 54–71. [CrossRef] [PubMed]
109. Wang, G.; Marcucci, G.; Peters, B.; Braidotti, M.C.; Muckli, L.; Faccio, D. Human-Centred Physical Neuromorphics with Visual Brain-Computer Interfaces. *Nat. Commun.* **2024**, *15*, 6393. [CrossRef] [PubMed]

110. Cooney, C.; Folli, R.; Coyle, D. Neurolinguistics Research Advancing Development of a Direct-Speech Brain-Computer Interface. *iScience* **2018**, *8*, 103–125. [CrossRef] [PubMed]
111. Müller-Putz, G.; Leeb, R.; Tangermann, M.; Höhne, J.; Kübler, A.; Cincotti, F.; Mattia, D.; Rupp, R.; Müller, K.-R.; Millán, J.d.R. Towards Noninvasive Hybrid Brain–Computer Interfaces: Framework, Practice, Clinical Application, and Beyond. *Proc. IEEE* **2015**, *103*, 926–943. [CrossRef]
112. Schirner, M.; Deco, G.; Ritter, P. Learning How Network Structure Shapes Decision-Making for Bio-Inspired Computing. *Nat. Commun.* **2023**, *14*, 2963. [CrossRef] [PubMed]
113. Angrick, M.; Luo, S.; Rabbani, Q.; Candrea, D.N.; Shah, S.; Milsap, G.W.; Anderson, W.S.; Gordon, C.R.; Rosenblatt, K.R.; Clawson, L.; et al. Online Speech Synthesis Using a Chronically Implanted Brain–Computer Interface in an Individual with ALS. *Sci. Rep.* **2024**, *14*, 9617. [CrossRef] [PubMed]
114. Shoaran, M.; Shin, U.; Shaeri, M. Intelligent Neural Interfaces: An Emerging Era in Neurotechnology. In Proceedings of the 2024 IEEE Custom Integrated Circuits Conference (CICC), Denver, CO, USA, 21–24 April 2024; pp. 1–7. [CrossRef]
115. Schalk, G.; Miller, K.J.; Anderson, N.R.; Wilson, J.A.; Smyth, M.D.; Ojemann, J.G.; Moran, D.W.; Wolpaw, J.R.; Leuthardt, E.C. Two-Dimensional Movement Control Using Electrocorticographic Signals in Humans. *J. Neural Eng.* **2008**, *5*, 75–84. [CrossRef]
116. Leuthardt, E.C.; Gaona, C.; Sharma, M.; Szrama, N.; Roland, J.; Freudenberg, Z.; Solis, J.; Breshears, J.; Schalk, G. Using the Electrocorticographic Speech Network to Control a Brain-Computer Interface in Humans. *J. Neural Eng.* **2011**, *8*, 036004. [CrossRef]
117. Mathis, M.W.; Perez Rotondo, A.; Chang, E.F.; Tolias, A.S.; Mathis, A. Decoding the Brain: From Neural Representations to Mechanistic Models. *Cell* **2024**, *187*, 5814–5832. [CrossRef]
118. Richards, B.A.; Lillicrap, T.P.; Beaudoin, P.; Bengio, Y.; Bogacz, R.; Christensen, A.; Clopath, C.; Costa, R.P.; de Berker, A.; Ganguli, S.; et al. A Deep Learning Framework for Neuroscience. *Nat. Neurosci.* **2019**, *22*, 1761–1770. [CrossRef] [PubMed]
119. Makin, J.G.; Moses, D.A.; Chang, E.F. Machine Translation of Cortical Activity to Text with an Encoder–Decoder Framework. *Nat. Neurosci.* **2020**, *23*, 575–582. [CrossRef]
120. Valle, G.; Katic Secerovic, N.; Eggemann, D.; Gorskii, O.; Pavlova, N.; Petrini, F.M.; Cvancara, P.; Stieglitz, T.; Musienko, P.; Bumbasirevic, M.; et al. Biomimetic Computer-to-Brain Communication Enhancing Naturalistic Touch Sensations via Peripheral Nerve Stimulation. *Nat. Commun.* **2024**, *15*, 1151. [CrossRef] [PubMed]
121. Duraivel, S.; Rahimpour, S.; Chiang, C.-H.; Trumpis, M.; Wang, C.; Barth, K.; Harward, S.C.; Lad, S.P.; Friedman, A.H.; Southwell, D.G.; et al. High-Resolution Neural Recordings Improve the Accuracy of Speech Decoding. *Nat. Commun.* **2023**, *14*, 6938. [CrossRef] [PubMed]
122. Hueber, P.; Tang, G.; Sifalakis, M.; Liaw, H.-P.; Micheli, A.; Tomen, N.; Liu, Y.-H. Benchmarking of Hardware-Efficient Real-Time Neural Decoding in Brain–Computer Interfaces. *Neuromorphic Comput. Eng.* **2024**, *4*, 24008. [CrossRef]
123. Farwell, L.A.; Donchin, E. Talking off the Top of Your Head: Toward a Mental Prosthesis Utilizing Event-Related Brain Potentials. *Electroencephalogr. Clin. Neurophysiol.* **1988**, *70*, 510–523. [CrossRef] [PubMed]
124. Wolpaw, J.R.; McFarland, D.J.; Neat, G.W.; Forneris, C.A. An EEG-Based Brain-Computer Interface for Cursor Control. *Electroencephalogr. Clin. Neurophysiol.* **1991**, *78*, 252–259. [CrossRef]
125. Crone, N.E.; Miglioretti, D.L.; Gordon, B.; Lesser, R.P. Functional Mapping of Human Sensorimotor Cortex with Electrocorticographic Spectral Analysis. II. Event-Related Synchronization in the Gamma Band. *Brain* **1998**, *121*, 2301–2315. [CrossRef] [PubMed]
126. Moses, D.A.; Leonard, M.K.; Chang, E.F. Real-Time Classification of Auditory Sentences Using Evoked Cortical Activity in Humans. *J. Neural Eng.* **2018**, *15*, 036005. [CrossRef]
127. Willett, F.R.; Kunz, E.M.; Fan, C.; Avansino, D.T.; Wilson, G.H.; Choi, E.Y.; Kamdar, F.; Glasser, M.F.; Hochberg, L.R.; Druckmann, S.; et al. A High-Performance Speech Neuroprosthesis. *Nature* **2023**, *620*, 1031–1036. [CrossRef] [PubMed]
128. Patrick-Krueger, K.M.; Burkhart, I.; Contreras-Vidal, J.L. The State of Clinical Trials of Implantable Brain–Computer Interfaces. *Nat. Rev. Bioeng.* **2024**, *2*, 1–18. [CrossRef]
129. Lee, A.T.; Chang, E.F.; Paredes, M.F.; Nowakowski, T.J. Large-Scale Neurophysiology and Single-Cell Profiling in Human Neuroscience. *Nature* **2024**, *630*, 587–595. [CrossRef]
130. Liu, S.; Wang, L.; Gao, R.X. Cognitive Neuroscience and Robotics: Advancements and Future Research Directions. *Robot. Comput. Integr. Manuf.* **2024**, *85*, 102610. [CrossRef]
131. Hbibi, B.; Khiari, C.; Wirsing, K.; Mili, L.; Baccar, K.; Mami, A. Identifying and Removing Interference and Artifacts in Multifractal Signals With Application to EEG Signals. *IEEE Access* **2023**, *11*, 119090–119105. [CrossRef]
132. Hong, N.; Kim, B.; Lee, J.; Choe, H.K.; Jin, K.H.; Kang, H. Machine Learning-Based High-Frequency Neuronal Spike Reconstruction from Low-Frequency and Low-Sampling-Rate Recordings. *Nat. Commun.* **2024**, *15*, 635. [CrossRef]
133. Stringer, C.; Pachitariu, M. Analysis Methods for Large-Scale Neuronal Recordings. *Science* **2024**, *386*, eadp7429. [CrossRef]
134. Speier, W.; Arnold, C.; Pouratian, N. Integrating Language Models into Classifiers for BCI Communication: A Review. *J. Neural Eng.* **2016**, *13*, 31002. [CrossRef] [PubMed]

135. Mitchell, P.; Lee, S.C.M.; Yoo, P.E.; Morokoff, A.; Sharma, R.P.; Williams, D.L.; MacIsaac, C.; Howard, M.E.; Irving, L.; Vrljic, I.; et al. Assessment of Safety of a Fully Implanted Endovascular Brain-Computer Interface for Severe Paralysis in 4 Patients: The Stentrode with Thought-Controlled Digital Switch (SWITCH) Study. *JAMA Neurol.* **2023**, *80*, 270–278. [CrossRef] [PubMed]
136. Mladenović, J. Standardization of Protocol Design for User Training in EEG-Based Brain–Computer Interface. *J. Neural Eng.* **2021**, *18*, 11003. [CrossRef]
137. Zhang, Z.; Chen, Y.; Zhao, X.; Fan, W.; Peng, D.; Li, T.; Zhao, L.; Fu, Y. A Review of Ethical Considerations for the Medical Applications of Brain-Computer Interfaces. *Cogn. Neurodyn.* **2024**, *18*, 3603–3614. [CrossRef]
138. Brocal, F. Brain-Computer Interfaces in Safety and Security Fields: Risks and Applications. *Saf. Sci.* **2023**, *160*, 106051. [CrossRef]
139. King, B.J.; Read, G.J.M.; Salmon, P.M. The Risks Associated with the Use of Brain-Computer Interfaces: A Systematic Review. *Int. J. Hum. Comput. Interact.* **2024**, *40*, 131–148. [CrossRef]
140. Awuah, W.A.; Ahluwalia, A.; Darko, K.; Sanker, V.; Tan, J.K.; Tenkorang, P.O.; Ben-Jaafar, A.; Ranganathan, S.; Aderinto, N.; Mehta, A.; et al. Bridging Minds and Machines: The Recent Advances of Brain-Computer Interfaces in Neurological and Neurosurgical Applications. *World Neurosurg.* **2024**, *189*, 138–153. [CrossRef] [PubMed]
141. Livanis, E.; Voultsos, P.; Vadikolias, K.; Pantazakos, P.; Tsaroucha, A. Understanding the Ethical Issues of Brain-Computer Interfaces (BCIs): A Blessing or the Beginning of a Dystopian Future? *Cureus* **2024**, *16*, 58243. [CrossRef]
142. Lu, Y.; Tian, H.; Cheng, J.; Zhu, F.; Liu, B.; Wei, S.; Ji, L.; Wang, Z.L. Decoding Lip Language Using Triboelectric Sensors with Deep Learning. *Nat. Commun.* **2022**, *13*, 1401. [CrossRef] [PubMed]
143. Min, B.; Ross, H.; Sulem, E.; Veyseh, A.P.B.; Nguyen, T.H.; Sainz, O.; Agirre, E.; Heintz, I.; Roth, D. Recent Advances in Natural Language Processing via Large Pre-Trained Language Models: A Survey. *ACM Comput. Surv.* **2023**, *56*, 1–40. [CrossRef]
144. Mehrish, A.; Majumder, N.; Bharadwaj, R.; Mihalcea, R.; Poria, S. A Review of Deep Learning Techniques for Speech Processing. *Inf. Fusion* **2023**, *99*, 101869. [CrossRef]